RSUS: A Novel Upsampling Layer for Semantic Segmentation Network of Remote Sensing Images

Yaning Liu, Ronghao Yang, Shaoda Li, Junxiang Tan, Guyue Hu, Chenghao Liu, Yunfan Luo, Zhiqiu Liang, and Tian Dong

***Abstract*—Semantic segmentation of remote sensing images (RSIs) often struggles with boundary blurring, structural discontinuity, and category confusion due to limitations in conventional interpolation and dynamic upsampling methods. This article proposes RSUS, a new upsampling layer designed to preserve semantic consistency and spatial structure. RSUS consists of three components: 1) global context vector aggregation (GCVA) for content-aware prediction kernels that introduce global priors into local feature reconstruction; 2) cross-scale anisotropic implicit positional encoding (CAIPE) for direction-sensitive spatial deformation and structural alignment; and 3) adaptive high-frequency structure gating (AHSG) to enhance boundary-related frequency responses. In addition, persistent homology (PH) is used as a topological analysis tool to assess connectivity and structure preservation beyond pixel-level metrics. Extensive experiments on five datasets (International Society for Photogrammetry and Remote Sensing (ISPRS) Vaihingen, ISPRS Potsdam, LoveDA, UAVid, and Xining) demonstrate that RSUS improves performance across mainstream segmentation networks, outperforming advanced upsampling methods. Analyses show RSUS alleviates structural misalignment, improves category consistency, and recovers fine-grained boundaries in remote sensing segmentation. The source code is available at: https://github.com/Ronin-711/RSUS**

***Index Terms*—Content-aware reconstruction, feature upsampling, persistent homology (PH), remote sensing image (RSI) semantic segmentation, structural preservation.**

I. Introduction

REMOTE sensing image (RSI) semantic segmentation aims to assign a semantic label to each pixel in an image and is one of the core tasks in the intelligent interpretation of remote sensing data [1]. Owing to its high efficiency and low cost, this technology has been widely applied in environmental monitoring [2], land cover classification [3], [4], geological environment investigation [5], and urban planning [6]. With the development of deep learning, the recognition accuracy of RSI semantic segmentation in complex scenes has been continuously improved. However, due to the significant complexity of different ground objects in terms of spatial organization, spectral response, and scale distribution, how to simultaneously maintain semantic consistency and spatial structural integrity remains a major challenge for this task.

In recent years, a large number of semantic segmentation models have been proposed for different types of RSIs and different ground object targets, including specialized methods for road extraction [7], land cover classification [8], and complex landform interpretation [9]. Existing studies mainly improve segmentation performance by optimizing backbone network architectures [10], modeling long-range contextual dependencies [11], and designing intraclass compensation mechanisms to alleviate intraclass variation while preserving spatial structural integrity [9], [12], [13]. Although these methods differ in network design and optimization strategies, they generally rely on upsampling operations to map low-resolution semantic features into high-resolution spatial predictions. Therefore, the upsampling layer is a key fundamental module in RSI semantic segmentation, and the quality of its feature reconstruction has an important impact on the final segmentation results [14].

Existing upsampling methods can be roughly divided into three categories. Bilinear interpolation and nearest-neighbor interpolation are the most commonly used nonparametric upsampling operators [9], [13], [15], [16], [17]. They reconstruct features based on the spatial distance between pixels and have the advantages of simple implementation and low computational cost. However, due to the lack of explicit modeling of semantic information, they are unable to accurately recover complex structures. Deconvolution (transposed convolution) methods [18] improve feature reconstruction to some extent by learning fixed upsampling kernels. However, because the convolution kernels are shared globally, they cannot be adaptively adjusted according to instance content, and large kernels also introduce additional computational burden. With the development of dynamic networks [19], content-adaptive upsampling methods have gradually shown greater potential. For example, CARAFE++ [15] achieves content-aware feature reconstruction by generating position-specific reassembly kernels, thereby improving representational capability while maintaining efficiency. Subsequently, researchers further proposed dynamic upsampling based on affinity modeling [20], encoder-guided dynamic upsampling [21], and guided upsampling based on point membership relationships [22]. Although these methods improve upsampling quality to a

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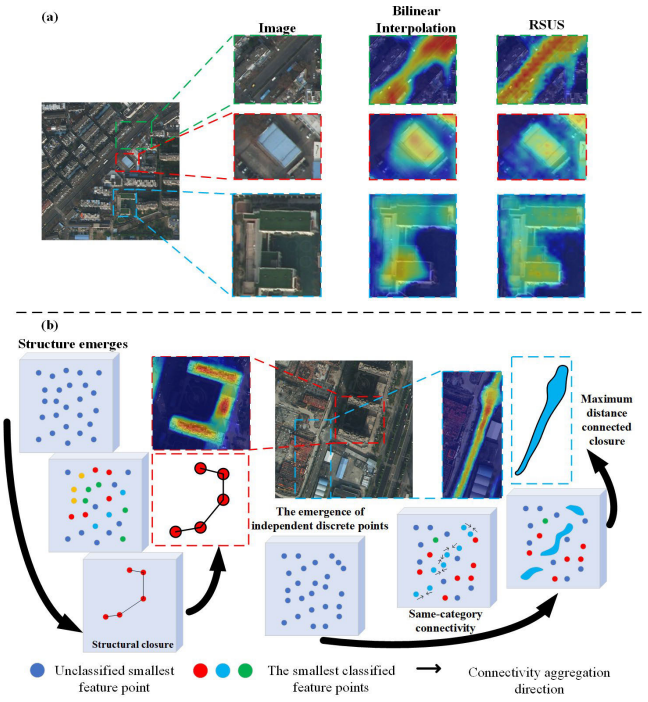


Fig. 1. Challenges in remote sensing semantic segmentation and Illustration of the PH mechanism. (a) Typical challenges of remote sensing semantic segmentation. (b) Visualization of the PH mechanism for remote sensing semantic segmentation.

certain extent, their ability to recover structural information remains insufficient in RSI scenarios.

This is because ground object categories in RSIs usually exhibit significant scale variation, complex spectral distributions, and strong spatial topological dependencies. As shown in Fig. 1(a), the same category may present markedly different spectral characteristics in different regions, while different categories may exhibit similar spectral responses. Meanwhile, ground objects such as roads and water systems usually possess continuous spatial structures and stable topological relationships. These characteristics make it difficult for upsampling methods that rely only on local neighborhood information to simultaneously ensure semantic consistency and spatial structural integrity. Especially during the upsampling stage, most existing methods still mainly rely on local information or fixed patterns for feature reconstruction and lack effective modeling of global structural information and category distribution patterns. As a result, they are prone to problems such as blurred boundaries, structural discontinuities, and category confusion in complex scenes, thereby limiting further improvement in model performance.

To address the above issues, this article proposes a content-aware upsampling method for RSI semantic segmentation, termed RSUS. The core idea of this method is to integrate global structural information with local detail constraints to achieve the collaborative recovery of semantic consistency and spatial structure. Specifically, RSUS generates dynamic prediction kernels through a global feature-guided mechanism to enhance intraclass feature consistency; meanwhile, it introduces cross-scale anisotropic spatial position encoding to constrain the spatial distribution of high-resolution features. In addition, a frequency-domain structural gating module is designed to strengthen the representation of high-frequency boundary information. Through these designs, RSUS can more effectively alleviate structural displacement during the upsampling process and improve the reconstruction capability for complex boundaries and fine-grained structures.

Furthermore, relying solely on traditional pixel-level evaluation metrics is insufficient to comprehensively reflect the structural distortions and category confusion introduced during the upsampling process. Such metrics mainly focus on matching accuracy at the pixel level, while failing to characterize category connectivity, spatial structure preservation, and the topological relationships among different categories. To this end, this article introduces persistent homology (PH) [23] as a supplementary evaluation tool to perform multiscale structural analysis of semantic segmentation results from a topological perspective. As shown in Fig. 1(b), PH describes the topological evolution of features across scales by constructing a Persistence Diagram [24], thereby characterizing the category connectivity and structure-preserving capability of the segmentation results. Compared with traditional pixel-level evaluation methods, PH can provide a more interpretable analytical perspective on the recovery of complex geographic structures and offer supplementary evidence for evaluating the effectiveness of the upsampling mechanism.

Specifically, the main contributions of this article are as follows.

1) Focusing on structural detail degradation in RSI semantic segmentation, this article analyzes the performance impact of upsampling modules and develops a multidimensional evaluation framework based on PH. Beyond traditional pixel-wise metrics, topological distance and Betti error are introduced to quantify object connectivity completeness, delivering a novel topologically interpretable perspective for measuring the structure-preserving capacity of upsampling operators.
2) We propose RSUS, a multibranch spatial-frequency collaborative upsampling layer built on content-aware reorganization for RSI semantic segmentation. Different from CARAFE++ and FADE, which only use local features for kernel prediction, RSUS exploits global semantic context to reduce intraclass inconsistency. Anisotropic implicit neural representation (INR) positional regularization and block-wise DCT gating are integrated to restrain structural deformation and enhance high-frequency edge responses, two components seldom adopted in existing upsampling designs. Einstein summation optimizes feature reconstruction to eliminate redundant tensor expansion and reduce GPU memory usage.
3) Extensive experiments on five multisource RSI datasets demonstrate that RSUS brings stable performance improvements when combined with six mainstream segmentation backbones, outperforming state-of-the-art (SOTA) upsampling methods. Ablation tests and topological evaluations validate the complementary capability of our dedicated modules to recover

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complete geographic edges and continuous ground object structures.

II. Related Works

A. Semantic Segmentation of RSIs

As a typical dense prediction task, semantic segmentation aims to assign a corresponding semantic category to each pixel in an image and is of great research value in tasks such as scene understanding, object recognition, and region interpretation [25]. Compared with natural image semantic segmentation, RSI semantic segmentation faces challenges such as more complex background interference, more significant intraclass variation, and more complicated spatial topological relationships [8], [16], [26]. The fundamental reason lies in the greater diversity of geospatial targets in RSIs, where different categories exhibit significant differences in scale, texture, shape, and distribution patterns. For example, tasks such as building extraction [27], [28], [29], road extraction [7], [30], farmland identification [31], [32], and geological hazard and complex landform interpretation [9], [33] all impose higher requirements on the semantic representation and structural recovery capabilities of models.

To address the complex characteristics of RSI segmentation tasks, existing studies have mainly improved segmentation performance from the perspectives of global–local information modeling, multidomain feature fusion, and long-range dependency modeling. For example, LOGCAN++ [13] enhances the model's ability to characterize contextual information by combining global perception and local perception strategies. SFFNet [34] improves the selection and representation of key features by jointly modeling spatial-domain and frequency-domain information. CM-UNet [35] strengthens the interaction between global and local features by introducing the Mamba mechanism. FarSeg++ [16] enhances the performance of high-resolution RSI semantic segmentation from the perspective of relation-, optimization-, and object-based foreground modeling. For hyperspectral image classification, existing approaches, including AF2GNN [36], [37], [38], introduce adaptive graph filters and aggregator fusion modules. More recently, adaptive homogeneous graph learning and self-supervised low-pass contrastive graph clustering methods have further investigated structure-aware filtering and graph representation learning tailored to hyperspectral imagery. SCSM [8] proposes a scene-coupled semantic mask network to reconstruct the attention mechanism, enabling it to better handle geospatial object segmentation tasks in complex scenes. In addition, the method of solving the coexistence of heterogeneous textures with uneven distribution in frequency components from the perspective of frequency domain has also been effectively applied to the task of RSI [39], [40], [41], [42], which confirms that the method of frequency domain is of great significance for the semantic segmentation task of RSI.

Although the above methods have achieved promising results in feature extraction and semantic modeling, most of them are still built upon the encoder–decoder framework and rely on traditional interpolation methods to perform feature upsampling during the decoding stage. Although such methods can restore feature resolution to a certain extent, their lack of adaptive modeling of target semantic attributes and global spatial relationships often makes it difficult to accurately recover the structural details and boundary information of geospatial targets in complex remote sensing scenes, leading to problems such as distorted semantic representation, boundary misalignment, and the loss of fine-grained structures after upsampling. Motivated by this, this article approaches the problem from the perspective of upsampling module design and proposes RSUS, an upsampler for RSI semantic segmentation that can directly replace traditional upsampling layers, so as to enhance the model's structural recovery and semantic representation capabilities in complex remote sensing scenes.

B. Feature Upsampling

The upsampling module is a key component in encoder–decoder networks for restoring spatial resolution and reconstructing target detail information. In current studies, nearest-neighbor interpolation and bilinear interpolation remain the most commonly used upsampling methods due to their simple implementation and low computational cost. However, traditional interpolation methods essentially rely on fixed kernel parameters and lack the ability to adaptively regulate the semantic content of input features, which makes them prone to detail blurring and structural distortion in complex scenes. To overcome this limitation, researchers have further proposed learnable upsampling methods, such as transposed convolution, to enhance feature reconstruction through learnable convolution kernels. However, due to the inherent unevenness of transposed convolution during the reconstruction process, it is not a strict inverse convolution operator and is likely to introduce artifacts, thereby affecting reconstruction quality. To address this issue, Huang et al. [18] proposed a new deconvolution scheme that is more suitable for network architectures requiring high-quality upsampling. Nevertheless, such deconvolution-based methods are usually accompanied by high computational cost, which limits their application in high-resolution vision tasks.

Against the background of balancing reconstruction quality and computational efficiency, content-aware upsampling methods have gradually become a major research focus. CARAFE [15] proposed a lightweight upsampling scheme based on content-aware reassembly kernels, which enables more flexible feature reconstruction at relatively low computational cost. Building upon CARAFE, Lu et al. [21] further proposed FADE, which improves the representational capability of upsampling results by dynamically balancing semantic information and detail information. Liu et al. [19], from the perspective of point sampling, proposed DySample to bypass the explicit generation of content-aware kernels and thereby improve upsampling efficiency. In addition to spatial-domain methods, FreqFusion [25] and MSAU [43] further considered the Nyquist–Shannon sampling theorem [44] and improved postupsampling detail recovery by modeling frequency-domain information and fusing components of different frequencies. Meanwhile, SAPA [22], based on point membership relationships, generates similarity- aware kernels by comparing the similarity between encoder feature points

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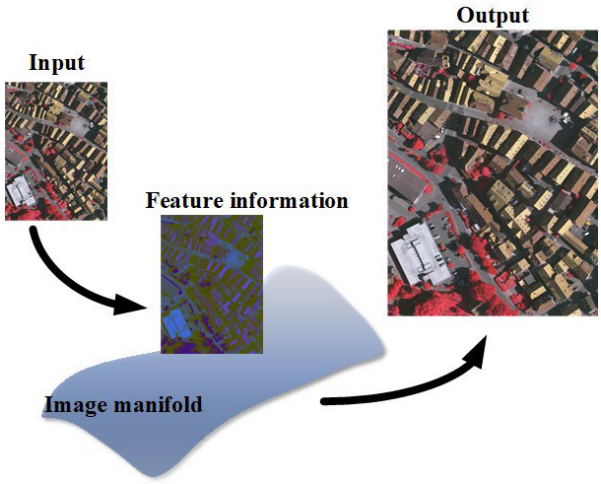


Fig. 2. Visualization of the manifold hypothesis.

and local decoder regions, thereby enhancing feature alignment capability. In addition, AnyUp [45], LoftUp [46], and JAFAR [47] realize upsampling based on window attention, coordinate attention, and spatial feature modulation strategies, respectively, further extending the application of attention mechanisms in feature reconstruction.

III. Methods

A. Motivation

In encoder–decoder-based RSI semantic segmentation models [48], [49], the upsampling operation is a key step in the decoder stage for restoring spatial resolution. Its goal is to recover low-resolution abstract semantic information into high-resolution feature representations with spatial structure.

Due to its advantages of requiring no additional parameters and incurring low computational cost, bilinear interpolation is adopted as the basic upsampling method in most existing RSI semantic segmentation models [9], [50], [51]. However, bilinear interpolation is essentially based on the local linearity assumption in Euclidean space and uses fixed weighting kernels to perform weighted averaging over neighboring pixels. This mechanism ignores the complex spatial heterogeneity of RSIs and is therefore inadequate for handling nonlinear semantic boundaries [15], [22]. From the perspective of feature representation, it is widely believed in machine learning that high-dimensional visual data often lie on low-dimensional manifold structures [52], [53]. Such manifolds encode not only appearance features such as texture and spectral information, but also implicit spatial structures and semantic relationships, as illustrated in Fig. 2. Since bilinear interpolation performs only arithmetic averaging in Euclidean space, the reconstructed feature points often deviate from the underlying data manifold, which is macroscopically manifested as blurred object boundaries and structural distortion.

From a representation learning perspective, remote sensing semantic features serve as constraints imposed on underlying low-dimensional structures. Accordingly, the upsampling module is required to jointly model content priors and structural priors, while interpolation relying on fixed Euclidean distance kernels fails to fully characterize such structural properties. In this article, we propose RSUS, a content-aware upsampling framework tailored for RSI, as illustrated in Fig. 3. SOTA upsampling paradigms, including FreqFusion, SAPA, and FADE, predominantly adopt frequency decomposition or feature reconstruction pipelines to prioritize spatial content recovery. By contrast, RSUS leverages a structurally constrained content-aware paradigm to achieve seamless integration of content and structural priors and addresses the inherent limitation of Euclidean interpolation in capturing low-dimensional structural patterns. Specifically, RSUS comprises three core modules: cross-scale anisotropic implicit position encoding, adaptive high-frequency structure gating (AHSG), and a content- aware prediction kernel built upon global context vector aggregation (GCVA). Cross-scale anisotropic implicit positional encoding (CAIPE) exploits position encoding to model anisotropic positional correlations among ground objects within remote sensing scenes, mitigating structural offsets during feature upsampling. Unlike classical position encoding schemes that only learn relative coordinate mappings [54], CAIPE learns orientation-specific sensitivity for each spatial position and projects local coordinates into an anisotropic geometric space. This design enables accurate modeling of positional dependencies for anisotropic ground targets without incurring heavy computational overhead for positional relation calculations. Similar to other advanced approaches, AHSG conducts feature construction in the frequency domain. Nevertheless, it differs from frequency domain upsampling methods such as FreqFusion by centering on frequency responses of structural features, which are embedded into the network as explicit structural constraints. This design bypasses cumbersome global feature modeling operations within the frequency domain. Rooted in global feature modeling principles, GCVA leverages global context aggregation to generate content-aware prediction kernels, overcoming the bottleneck of existing content-aware operators that only exploit local feature information.

Benefiting from the joint collaboration of the above three modules, the proposed RSUS effectively mitigates structural distortion and category misclassification artifacts arising from conventional feature upsampling strategies.

B. Overview of RSUS

RSUS is designed for structure-aware upsampling in remote sensing segmentation. It consists of a GCVA-based content-aware main branch, an AHSG frequency compensation branch, and a CAIPE structural prior branch. Specifically, the GCVA main branch is responsible for generating content-aware prediction kernels that integrate global contextual information, the AHSG branch is used to compensate for boundary-related high-frequency structural responses, and the CAIPE branch provides cross-scale structural positional priors.

As shown in Fig. 3, RSUS consists of a content-aware prediction kernel based on GCVA, CAIPE, and AHSG. Specifically, given an input feature $F_i \in \mathbb{R}^{C\times H\times W}$ and a target upsampling factor s, the goal of RSUS is to produce a high-scale feature $F_o \in \mathbb{R}^{C\times H*s\times W*s}$ with rich geometric details.

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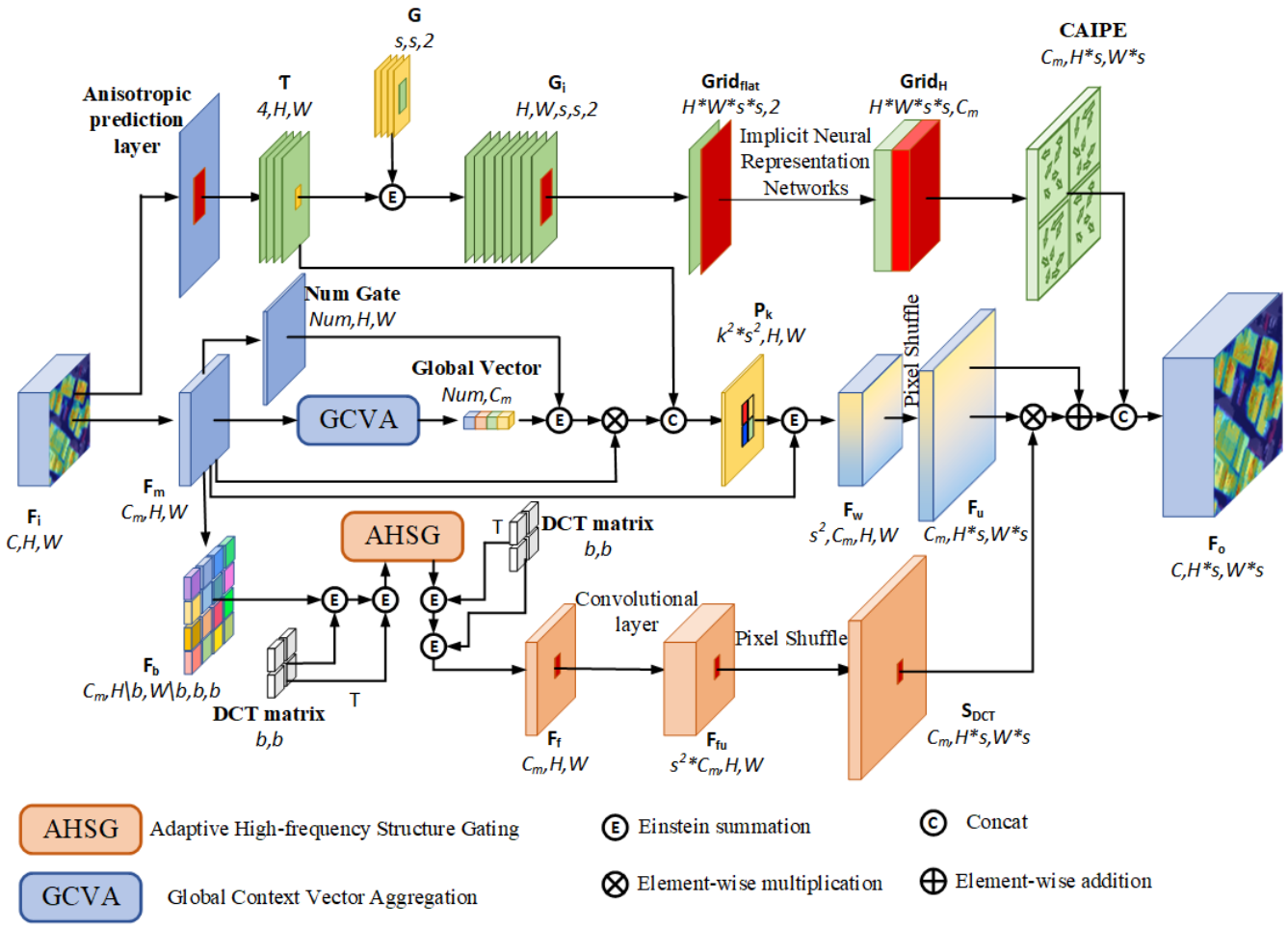


Fig. 3. Overall architecture of the RSUS, consisting of a main content-aware module with global context feature aggregation, an adaptive high-frequency structural gating branch, and a cross-scale anisotropic position encoding branch.

This process not only improves the spatial resolution but also preserves the topological consistency of feature reconstruction in the nonlinear manifold space.

In terms of the processing pipeline, RSUS adopts a dual-branch parallel architecture. The main branch first injects global semantic information into low-dimensional spatial manifold features through GCVA, and then generates content-aware prediction kernels to perform reassembly sampling on the original features. Subsequently, AHSG is employed to enhance high- frequency boundary information in the frequency domain. The auxiliary branch is responsible for extracting anisotropic features of individual pixels and generating CAIPE to provide explicit spatial structural constraints. Finally, the enhanced features produced by the main branch are concatenated with the positional encoding generated by the auxiliary branch along the channel dimension to obtain the final upsampled output feature.

The overall formulation of RSUS is given in the following equation:

$$\begin{aligned} F_o &= \mathcal{F}_{RSUS}(F_i, s) \\ &= \Phi\left(\mathcal{R}\left(\mathcal{G}\left(F_i, s\right) \otimes M_{dct}\left(F_i, s\right)\right) \| P_{inr}(F_i, s)\right) \end{aligned} \tag{1}$$

where $\mathcal{F}_{RSUS}$ denotes the overall pipeline of RSUS, Φ denotes linear mapping, $\mathcal{R}(\cdot)$ denotes residual refinement combined with high-frequency structural gating, $\mathcal{G}(\cdot)$ denotes the GCVA and reassembly sampling branch, $M_{dct}(\cdot)$ denotes the adaptive high-frequency gating matrix branch generated by AHSG, $P_{inr}(\cdot)$ denotes the branch for generating CAIPE, and $\otimes$ and $\|$ denote the Hadamard product and channel-wise concatenation, respectively.

C. Anisotropic Implicit Spatial Location Encoding Across Scales

The discrete pixel-grid mapping of upsampled features tends to induce local geometric distortions in the feature space, which manifest as edge deformation and nonrigid displacement of ground objects in high-resolution reconstruction results. To address this issue, we introduce positional encoding [54] into the feature upsampling process and propose a CAIPE module. Unlike conventional position encoding methods that merely model general spatial relationships, CAIPE aims to establish an implicit spatial structure constraint mechanism for direction-sensitive deformation modeling in upsampling tasks. It achieves this by characterizing the geometric anisotropy of local regions within a continuous coordinate manifold. The generation of CAIPE follows a cascaded transformation process from discrete representation to continuous manifold, and then to high-dimensional feature space. Specifically, the

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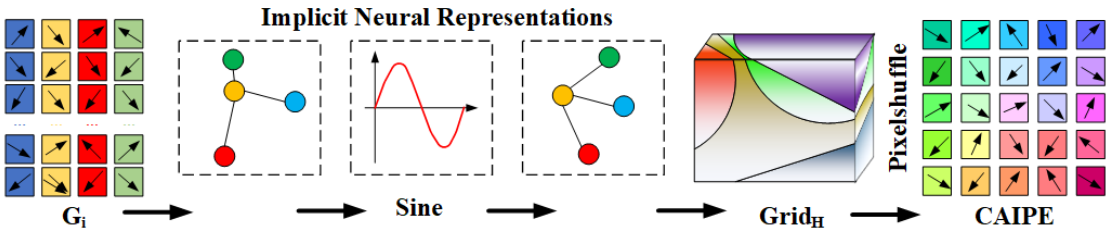


Fig. 4. Projection of low-resolution pixel coordinate information to high-resolution space via an INR network.

input feature $F_i \in \mathbb{R}^{C\times H\times W}$ is first fed into a lightweight predictor composed of convolutional layers to estimate the local geometric transformation parameters $\mathcal{T} \in \mathbb{R}^{4\times H\times W}$. At each spatial location (h, w), the four channels of $\mathcal{T}$ are mapped to predict a 2×2 local transformation matrix

$$M = \begin{bmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{bmatrix}$$

which is used to model direction-related scaling and axis-coupled deformation trends. The formulas for estimating the local geometric transformation parameters $\mathcal{T}$ and reconstructing the second-order affine transformation matrix M are given in the following equations:

$$\mathcal{T} = \vartheta(F_i) \tag{2}$$

$$M = \mathcal{PR}(\mathcal{T}) \tag{3}$$

where $\vartheta(\cdot)$ denotes the lightweight predictor, and $\mathcal{PR}$ denotes the mapping from the channel semantic space to the affine transformation space. Specifically, for $\mathcal{T}$, the 4-D.feature vector $t_{h,w} \in \mathbb{R}^4$ at the designated spatial position (h, w) of a specific sample is isomorphically mapped to a local feature mapping $m_{h,w} \in \mathbb{R}^{2\times2}$ with complete affine transformation components at position (h, w) in matrix M, so that it can act on the local coordinate grid in the form of an operator. The isomorphic mapping process can be expressed as $m_{h,w} = mat(t_{h,w})$, where $mat(\cdot)$ denotes the isomorphic mapping of $\mathbb{R}^4 \to \mathbb{R}^{2\times2}$.

To realize the adaptive evolution of the coordinate space, a normalized local subpixel coordinate grid $G \in [-1, 1]^{s\times s\times2}$ is introduced and combined with M under the Einstein summation convention [55]. In this way, the predicted transformation matrix is applied to each local grid point, so that spatial relative positional relationships are established among the anisotropic features of individual pixels, yielding the anisotropic coordinate transformation matrix $G_i \in \mathbb{R}^{H\times W\times s\times s\times2}$. As a result, the model can dynamically adjust the scaling and deflection of the receptive field according to the morphological characteristics of remote sensing targets, thereby constructing a structure-aware nonlinear manifold in the feature space. The anisotropic coordinate transformation matrix G_i can be written as follows:

$$G_i = Einsum(M, G) \tag{4}$$

where *Einsum* denotes the standard Einstein summation convention.

Since the local coordinates of the transformation matrix G_i are still discrete and resolution-dependent geometric descriptions, an INR is introduced to transform them into cross-scale shared and continuously queryable structural priors. Specifically, the deformed geometric coordinates $\text{Grid}_{flat} \in \mathbb{R}^{H*W*s^2\times2}$ of G_i are mapped into a high-dimensional structural feature space, yielding high-dimensional manifold features $Grid_H \in \mathbb{R}^{H*W*s^2,C_m}$, which are finally reorganized into high-resolution CAIPE through PixelShuffle [56]. As a cross-scale structural positional prior, CAIPE is used to constrain the spatial alignment of the initial features F_u sampled by the content-aware prediction kernel. In the INR, the core layer adopts the sinusoidal activation function, Sine. Owing to its infinitely differentiable property, the network can preserve high-frequency harmonic information in the input coordinates to a large extent. This implies that after the transformation by the anisotropic matrix M, even when the coordinates undergo slight rotation or stretching, the sinusoidal activation helps preserve high-frequency coordinate variations and supports smooth, continuous mapping. As shown in Fig. 4, by transforming the discrete feature reorganization problem into a query process over continuous functions, CAIPE achieves resolution-independent modeling capability. Even under different upsampling factors s, the INR can still provide consistent and smooth structural feature projection on the deformed G_i, thereby significantly enhancing the generalization ability of the operator in cross-scale scenarios. Therefore, the $P_{inr}(\cdot)$ structure in (1) can be further specified by the following equations:

$$CAIPE = P_{inr}(F_i, s) \tag{5}$$

$$CAIPE = PixelShuffle(W_2 \cdot \text{Sin}(W_1 \cdot G_i + b_1) + b_2) \tag{6}$$

where W and b denote the learnable weight matrix and bias term, respectively, and $Sin(\cdot)$ denotes the sinusoidal activation function.

D. Adaptive High-Frequency Structure Gating

In RSI semantic segmentation, since the high-frequency spectra associated with the boundaries of different categories vary [57], the enhancement of high-frequency structural responses is an important factor affecting RSI semantic segmentation performance [58]. According to the Nyquist–Shannon sampling theorem [44], frequency components higher than the Nyquist frequency, which is equal to half of the sampling rate, are permanently lost during downsampling. Therefore, for an upsampling layer that restores low-resolution semantic features into high-resolution spatial features, whether it can effectively enhance category-boundary high-frequency responses is an important criterion for evaluating its performance. To this end, we design an AHSG

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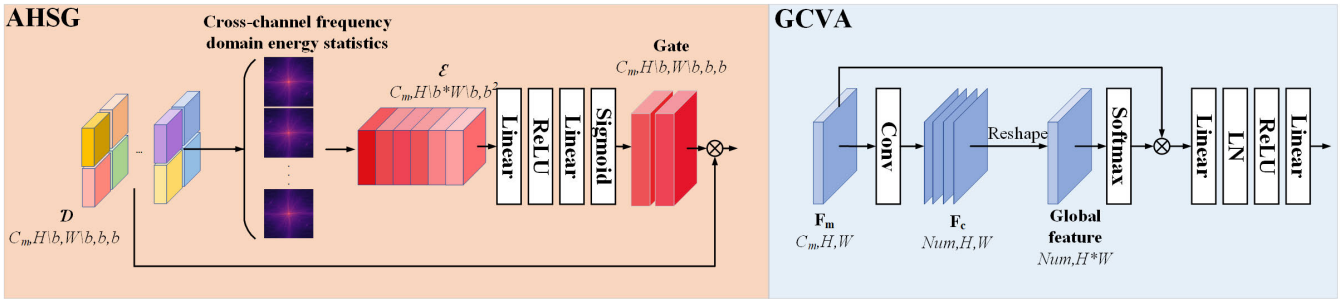


Fig. 5. Architectures of the AHSG and GCVA modules.

module in the frequency domain. AHSG does not attempt to reconstruct the true high-frequency components that have been irreversibly lost; instead, it adaptively adjusts the responses of different frequency components through local frequency-domain energy statistics, thereby enhancing the representation of boundary-related high-frequency information during the upsampling process.

As shown in Fig. 3, the low-dimensional manifold feature F_m is divided into b nonoverlapping local region features $F_b \in \mathbb{R}^{C_m \times (H\backslash b) \times (W\backslash b) \times b \times b}$ of the same size. This is because boundaries and textures in remote sensing scenes exhibit spatial nonstationarity; therefore, block-wise DCT is adopted to model local frequency-domain patterns, so as to avoid the over-smoothing of local structures caused by global spectrum analysis. Specifically, as can be seen from the following equations, F_b is subjected to Einstein summation with the standard discrete cosine transform matrix $D \in \mathbb{R}^{N \times N}$ and its transposed matrix, respectively, yielding the coefficient matrix $\mathcal{D} \in \mathbb{R}^{C_m \times (H\backslash b) \times (W\backslash b) \times b \times b}$ decomposed from local low-frequency to high-frequency components:

$$D = \begin{cases} \sqrt{\frac{1}{N}}, & \text{if } k = 0 \\ \sqrt{\frac{2}{N}} \cos\left(\frac{\pi(n+0.5)k}{N}\right) & \text{if } k > 0 \end{cases} \tag{7}$$

$$\mathcal{D} = Einsum\left(Einsum\left(D, F_b\right), D^T\right) \tag{8}$$

where in the standard discrete cosine transform matrix, N denotes the size of the orthogonal matrix, k denotes the row index, i.e., the frequency index, $k = 0$ denotes the DC component, $k > 0$ denotes the AC component, n denotes the column index, and D and D^T denote the standard discrete cosine transform matrix and its transpose, respectively.

As shown in the AHSG part of Fig. 5, cross-channel frequency-domain energy statistics are performed on the coefficient matrix $\mathcal{D}$. Specifically, the energy feature $\mathcal{E} \in \mathbb{R}^{C_m \times (H\backslash b) \times (W\backslash b) \times b^2}$ is obtained by computing the mean absolute value of the frequency components along the channel dimension, which represents the intensity of different frequency bands in the local region. Then, $\mathcal{E}$ is fed into an MLP composed of linear layers and mapped into a weight matrix $Gate \in \mathbb{R}^{C_m \times (H\backslash b) \times (W\backslash b) \times b \times b}$. This weight matrix is applied to $\mathcal{D}$, and inverse discrete cosine transformation is then performed to obtain the low-resolution gating feature $F_f \in \mathbb{R}^{C_m \times H \times W}$. Subsequently, F_f is projected by a convolution layer into a high-dimensional gating feature $F_{fu} \in \mathbb{R}^{(s^2 * C_m) \times H \times W}$, and *PixelShuffle* is applied to obtain the high-resolution DCT structural gate $S_{DCT} \in \mathbb{R}^{C_m \times H*s \times W*s}$. Therefore, $M_{dct}(\cdot)$ in (1) can be further specified by the following equations:

$$S_{DCT} = M_{dct}(F_i, s) \tag{9}$$

$$\mathcal{E} = \frac{1}{C} \sum_{c=1}^{C} |\mathcal{D}| \tag{10}$$

$$Gate = \sigma\left(W_2 \cdot \mathcal{R}eLU\left(W_1 \cdot vec\left(\mathcal{E}\right) + b_1\right) + b_2\right) \tag{11}$$

$$F_f = Einsum\left(Einsum\left(D^T, \mathcal{D} \otimes reshape\left(Gate\right)\right), D\right) \tag{12}$$

$$S_{DCT} = PixelShuffle\left(Conv\left(F_f\right)\right) \tag{13}$$

where C denotes the number of channels, σ denotes the Sigmoid activation function, $W_1 \in \mathbb{R}^{d \times b^2}$ and $W_2 \in \mathbb{R}^{b^2 \times d}$ denote the learnable weight matrices, d denotes the hidden-layer dimension, $vec(\cdot)$ denotes the operation of flattening $b \times b$ energy blocks into a b^2-dimensional vector, and b_1 and b_2 denotes the bias term.

E. Content-Aware Based on the Global Context

The content-aware prediction kernel can effectively adapt to local category-specific features and capture fine-grained variations within local regions. However, for RSIs containing spatially distributed instances of the same category, such kernels are limited in modeling long-range dependencies, leading to insufficient fusion of global contextual information [13]. Without global guidance, prediction kernels tend to rely on local feature similarity, which may result in intraclass feature collapse during large-scale upsampling and consequently cause intraclass inconsistency in the reconstructed results. In addition, existing content-aware reorganization methods typically involve the expansion of high-dimensional feature tensors, which introduces additional computational overhead and reduces inference efficiency [15], [21]. To address these limitations, we propose a global context-guided content-aware prediction kernel, which integrates global features to enhance long-range dependency modeling while redesigning the feature reorganization process to reduce redundant tensor expansion and improve computational efficiency.

As shown in the GCVA part of Fig. 5, GCVA aims to achieve long-range semantic compensation through global semantic prototypes, following the process of "semantic aggregation–nonlinear modulation–spatial reprojection." Specifically, GCVA uses lightweight convolutional layers to extract a set of global semantic prototypes $F_c \in \mathbb{R}^{Num \times H \times W}$

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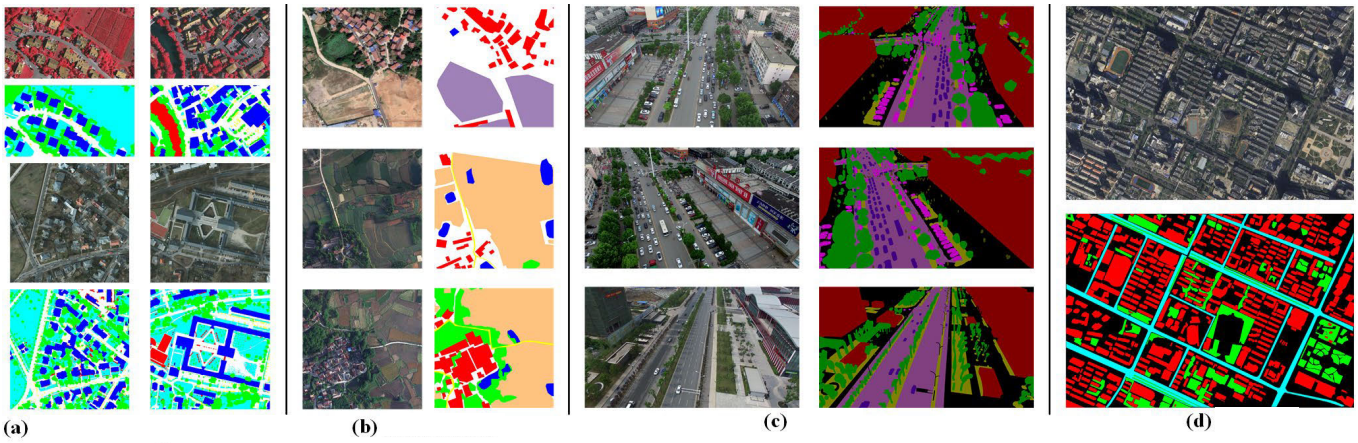


Fig. 6. Sample images and corresponding labels from four training datasets. (a) ISPRS Vaihingen and Potsdam. (b) LoveDA 2021. (c) UAVid. (d) Xining datasets.

from the input low-dimensional spatial manifold feature $F_m \in \mathbb{R}^{C_m \times H \times W}$. Here, *Num* denotes the number of target categories; therefore, the channel dimension of F_c corresponds to global semantic priors for different target categories. Then, F_c is reshaped into global contextual features, with the reshaping process given by $\mathbb{R}^{Num \times H \times W} \rightarrow \mathbb{R}^{Num \times H*W}$. A Softmax mapping is performed along the second dimension of the reshaped global contextual features and weighted onto F_m, thereby aggregating the global information of different categories with category-specific weights. Subsequently, the weighted features are passed through a linear layer to map category features in the spatial context vector $GlobalVector \in \mathbb{R}^{Num \times C_m}$ that contains all categories. Therefore, the process by which GCVA generates the global context vector can be expressed by the following equation:

$$GlobalVector = W\left(\psi^2\left(flat\left(Conv\left(F_m\right)\right)\right) \otimes F_m\right) + b \tag{14}$$

where W and b denote the learnable weight matrix and bias term, respectively. ψ^2 denotes performing the softmax mapping along the second dimension, and $flat(\cdot)$ denotes the reshaping process $\mathbb{R}^{Num \times H \times W} \rightarrow \mathbb{R}^{Num \times H*W}$.

The low-dimensional spatial manifold feature F_m is mapped by a convolutional layer parallel to GCVA into a gating representation $NumGate \in \mathbb{R}^{Num \times H \times W}$ that covers each category feature. Through Einstein summation with *GlobalVector*, this gating representation assigns the semantic prototypes of the global context vector to different local regions of different categories, and then uses them as category semantic priors to perform the Hadamard product with F_m. Subsequently, this global prior is concatenated with the anisotropic structural prior $\mathcal{T}$ to generate the final content-aware prediction kernel $P_k \in \mathbb{R}^{(k^2*s^2) \times H \times W}$. The prediction kernel P_k is then subjected to Einstein summation with F_m to obtain $F_w \in \mathbb{R}^{s^2 \times C_m \times H \times W}$, thereby enabling the perception kernel aggregated from the global context vector to perform prior prediction on the original features. Finally, F_w is reorganized by PixelShuffle to obtain the initial feature $F_u \in \mathbb{R}^{C_{mm} \times H*s \times W*s}$ sampled by the content-aware prediction kernel. Therefore, the process of generating F_u from $\mathcal{G}(\cdot)$ in (1) can be further specified by the following equations:

$$F_u = \mathcal{G}(F_i, s) \tag{15}$$

$$NumGate = Conv(F_m) \tag{16}$$

$$P_k = Einsum(NumGate, GlobalVector) \otimes F_m \| \mathcal{T} \tag{17}$$

$$F_u = PixelShuffle(Einsum(P_k, F_m)). \tag{18}$$

IV. Experiments

A. Datasets

To verify the generalization ability of RSUS across different sensors and different geographic regions, experiments are conducted on four public datasets and one self-constructed dataset. The public datasets include three aerial image datasets, namely International Society for Photogrammetry and Remote Sensing (ISPRS) Vaihingen [59], ISPRS Potsdam [59], and LoveDA [60], as well as one UAV image dataset, UAVid [61]. The self-constructed dataset is the aerial image dataset Xining. As shown in Fig. 6, examples of the original training images and their corresponding category labels from the five datasets are presented.

1) ISPRS Vaihingen: The Vaihingen dataset was released by the ISPRS in 2014, and the study area is located in the town of Vaihingen, Germany. Provided by the German Aerospace Center (DLR), it is a classic dataset for remote sensing semantic segmentation. The dataset contains 33 images with a ground sampling distance of 9 cm, together with their corresponding ground-truth labels, and the image sizes range from 1996×1995 to 3816×2550. Each orthophoto includes three multispectral bands (near-infrared, red, and green), one digital surface model (DSM), and one normalized DSM (nDSM). The Vaihingen dataset contains six categories: impervious surfaces, buildings, low vegetation, trees, cars, and clutter/background. In this article, the first five categories are selected for experiments. Meanwhile, standard false-color orthophotos composed of multispectral bands are used in the experiments. The training set consists of 16 images, namely images numbered 1, 3, 5, 7, 11, 13, 15, 17, 21, 23, 26, 28, 30, 32, 34, and 37, while the remaining 17 images are used for testing. In addition, to ensure training stability, the Vaihingen dataset is cropped into image patches of size 512×512.

2) ISPRS Potsdam: The Potsdam dataset, also provided by DLR, is another commonly used dataset for remote sensing semantic segmentation. It is located in Potsdam,

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Germany, and contains 38 aerial images together with their corresponding ground-truth labels, with a ground sampling distance of 5 cm. Each image has a size of 6000 × 6000. The dataset includes four multispectral bands (near-infrared, red, green, and blue), one DSM, and one nDSM. The label categories are the same as those in the Vaihingen dataset, and in this article, the first five categories are also selected for experiments. Meanwhile, true-color images composed of the red, green, and blue bands are used in the experiments. The training regions include images numbered 2_10, 2_11, 2_12, 3_10, 3_11, 3_12, 4_10, 4_11, 4_12, 5_10, 5_11, 5_12, 6_7, 6_8, 6_9, 6_10, 6_11, 6_12, 7_7, 7_8, 7_9, 7_10, 7_11, and 7_12. The remaining 14 images are used for testing. In addition, the Potsdam dataset is also cropped into patches of size 512 × 512 for the experiments.

3) LoveDA: The LoveDA dataset was introduced in 2021 as part of a semantic segmentation challenge organized by Wuhan University. It is a classic and challenging remote sensing semantic segmentation dataset, well known for its complex backgrounds and rich diversity in object scales and categories. The dataset contains 5987 images of size 1024 × 1024, including 2522 training images, 1669 validation images, and 1796 test images. It covers urban and rural areas in three cities in China, namely Nanjing, Changzhou, and Wuhan, with a total coverage area of 536.15 km^2 and a ground sampling distance of 0.3 m. LoveDA contains seven categories: building, road, water, vegetation, car, barren, and background. In particular, this article adopts the official online evaluation protocol of the LoveDA challenge.

4) UAVid: The UAVid dataset is a public UAV-platform semantic segmentation dataset proposed in 2020. Unlike aerial imaging platforms, it adopts an oblique imaging perspective, which poses additional challenges for remote sensing semantic segmentation models. The UAVid dataset was acquired at a UAV flight altitude of approximately 50 m and contains 820 images with resolutions of 4096 × 2160 or 3840 × 2160, including 600 training images, 70 validation images, and 150 test images. UAVid contains eight categories: background, building, road, tree, low vegetation, moving car, static car, and human. To ensure training stability, the UAVid dataset is cropped into patches of size 512 × 512 for the experiments.

5) Xining: To further verify the performance of RSUS under more complex urban geographical distribution, we use Beijing-3 RSIs to construct the Xining dataset, which covers the urban area of Xining, Qinghai, with a spatial resolution of 0.3 m. Xining has a unique geospatial structure, which is located in the typical faulted basin area of the northeastern margin of the Qinghai-Tibet Plateau, showing the characteristics of plateau valley urban geomorphology. In addition, it is characterized by dense buildings and rivers flowing through the city. Therefore, the Xining dataset is a challenging dataset that can be used to evaluate the resilience of complex geospatial topologies. The Xining dataset contains 1260 images of 1024 × 1024 size, including 1060 training images and 200 test images. It contains a total of seven classes: background, building, green space, water, road, wasteland, and farmland. The first five categories are used for experiments in this article. A detailed description of this dataset can be found in the Appendix.

B. Evaluation Methodology

In this section, we introduce the evaluation methods used to assess the performance of RSUS. We mainly employ two types of evaluation criteria, namely pixel-level metrics and topological analysis.

1) Pixel-Level Metrics: Pixel-level evaluation metrics are commonly used criteria for measuring the performance of semantic segmentation models. In this article, three mainstream pixel-level evaluation metrics are adopted, namely mean intersection over union (mIoU), mean $F1$ score (m$F1$), and overall accuracy (OA).

The formulas of these three evaluation metrics can be expressed as follows:

$$mIoU = \frac{1}{k}\sum_{k=1}^{k}\frac{TP_k}{TP_k + FP_k + FN_k} \tag{19}$$

$$P_k = \frac{TP_k}{TP_k + FP_k} \tag{20}$$

$$R_k = \frac{TP_k}{TP_k + FN_k} \tag{21}$$

$$mF1 = \frac{1}{k}\sum_{k=1}^{k}\frac{2 \times P_k \times R_k}{P_k + R_k} \tag{22}$$

$$OA = \frac{\sum_{k=1}^{k} TP_k}{\sum_{k=1}^{k} TP_k + FN_k} \tag{23}$$

where k denotes the total number of categories, TP_k (true positive) denotes the correct prediction of positive samples for the kth category, FN_k (false negative) denotes the positive samples of the kth category that are incorrectly classified as negative, FP_k (false positive) denotes the negative samples of the kth category that are incorrectly classified as positive, and TN_k (true negative) denotes the correct prediction of negative samples for the kth category.

2) Topological Analysis: To verify the recovery capability of RSUS for category connectivity and structural integrity during the upsampling process, we introduce PH from topological analysis as a supplementary tool to assist traditional metrics in evaluating the ability of RSUS to recover topological structures. PH is a method for measuring the robustness of features in data across different topological scales and is an important component of topological data analysis. It has been applied in multiple fields, including neuroscience [62], phase transition studies, and image artifact measurement [63]. PH typically characterizes the evolution of topological structures across different scales by constructing a filtration process. Its essence lies in describing topological invariants based on homology groups, such as connected components and hole structures. In RSI semantic segmentation tasks, these topological features can be used to characterize the connectivity and structural integrity of target regions, thereby providing a more reliable basis for evaluating the model's ability to recover spatial structures during the upsampling process.

Considering the RSIs addressed in this article, suppose that the output $\mathcal{P}(x, y)$ of the semantic segmentation network

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represents the predicted probability of a certain category at coordinate (x, y), which can be expressed by the following equation:

$$\mathcal{P}(x,y):\Omega \subset \mathbb{R}^2 \rightarrow [0,1] \tag{24}$$

$$\mathcal{K}_\varepsilon = \{(x,y) \in \Omega \,|\mathcal{P}\,(x,y) \geq \varepsilon\} \tag{25}$$

where Ω denotes the 2-D image domain, ε denotes the prediction threshold of the target category, and $\mathcal{K}_\varepsilon$ denotes the set of pixels corresponding to the binary image under different thresholds. Therefore, they form a nested sequence, as shown in the following equation:

$$\mathcal{K}_{\varepsilon 1} \subseteq \mathcal{K}_{\varepsilon 2}, \quad \varepsilon 1 > \varepsilon 2. \tag{26}$$

For each threshold ε, we define the homology groups and their corresponding Betti numbers, as given in the following equations:

$$H_k\left(\mathcal{K}_\varepsilon\right), \quad k = 0, 1 \tag{27}$$

$$\mathcal{B}_k\left(\varepsilon\right) = rank(H_k(\mathcal{K}_\varepsilon)) \tag{28}$$

where $H_0(\mathcal{K}_\varepsilon)$ denotes the 0-D homology, which represents the equivalence classes of connected components. $\mathcal{B}_0$ denotes the Betti number of the 0-D homology, specifically the number of connected components; in RSIs, it reflects the connected merging of pixels belonging to the same category and is computed as the rank of the 0-D homology group. $H_1(\mathcal{K}_\varepsilon)$ denotes the 1-D homology, which represents the persistence of loop structures across different scales and can reflect the model's ability to capture large-scale topological structures. $\mathcal{B}_1$ denotes the Betti number of the 1-D homology, specifically the number of mutually independent topological loop structures, and is computed as the rank of the 1-D homology group.

Based on the filtration and homology definitions in (24)–(28), we further introduce quantitative topology-consistency metrics to support the qualitative PH visualization. Specifically, for each semantic category, we compute the Bottleneck distance, Wasserstein distance, and Betti error between the predicted result and the corresponding ground truth. Specifically, it can be defined as follows:

$$d_B^k(c) = \inf_{\pi} \sup_{p \in PD_k(\hat{Y}_c)} \|p - \pi(p)\|_\infty, \quad k \in \{0,1\} \tag{29}$$

$$d_W^k(c) = \left(\inf_{\pi} \sum_{p \in PD_k(\hat{Y}_c)} \|p - \pi\|_2^r\right)^{1/r}, \quad k \in \{0,1\} \tag{30}$$

$$E_{\mathcal{B}_k} = \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} \left|\mathcal{B}_k(\hat{Y}_c) - \mathcal{B}_k(Y_c)\right|, \quad k \in \{0,1\} \tag{31}$$

where c denotes the semantic category index, and $k \in \{0,1\}$ denotes the homology dimension, with $k = 0$ corresponding to connected components and $k = 1$ corresponding to loop structures. $\hat{Y}_c$ and Y_c denote the predicted and ground-truth binary masks of class c, respectively. $PD_k(\hat{Y}_c)$ and $PD_k(Y_c)$ denote the k-dimensional persistence diagrams generated from $\hat{Y}_c$ and Y_c, respectively. p denotes a persistence point in $PD_k(\hat{Y}_c)$, and π denotes a feasible matching between $PD_k(\hat{Y}_c)$ and $PD_k(Y_c)$, where points are allowed to be matched to the birth-death diagonal. $\pi(p)$ denotes the point matched with p under π. $\|\cdot\|_\infty$ and $\|\cdot\|_2$ denote the infinity norm and Euclidean norm, respectively. The operator sup computes the maximum matching distance under a given matching, while $\inf_\pi$ selects the matching that minimizes this maximum distance over all feasible matchings. In the Wasserstein distance, r denotes the order of the distance. $\mathcal{C}$ denotes the set of semantic categories, and $\mathcal{B}_k(\cdot)$ denotes the kth Betti number.

The bottleneck distance measures the largest topological discrepancy between the predicted and ground-truth persistence diagrams, the Wasserstein distance measures their accumulated topological discrepancy, and the Betti error measures the difference in the number of connected components or loop structures.

C. Implementation Details

The experiments of RSUS are implemented based on PyTorch 2.1.0 under the MMSegmentation 1.2.2 framework [64]. All experiments are conducted on a single NVIDIA RTX 4090 GPU with 48 GB of memory, running Linux Ubuntu 24.04.3. In the training settings, all experiments use the AdamW optimizer, with an initial learning rate of 1×10^{-4}, momentum parameters $\beta_1 = 0.9$ and $\beta_2 = 0.999$, and a weight decay of 1×10^{-4}. For experiments with an input size of 512×512, the batch size is set to 16, while for experiments with an input size of 1024×1024, the batch size is set to 8. The total number of training epochs for all experiments is 150.

D. Experimental Results and Analysis

This section presents the experimental results of RSUS. For the RSI semantic segmentation task, we design two types of experiments: 1) effectiveness and universality validation experiments, which aim to evaluate the performance of RSUS when replacing the original upsampling layers in multiple advanced semantic segmentation models and 2) comparative experiments with advanced upsampling methods, which aim to examine whether RSUS can achieve superior performance in RSI semantic segmentation when compared with different SOTA upsampling schemes. Meanwhile, Section IV-D3 further employs visualizations of class feature distribution and topological analysis to complement the above experiments, so as to verify the clustering enhancement and structural recovery capabilities of RSUS. In addition, Section IV-D4 presents ablation experiments to evaluate the effectiveness of each component of RSUS.

1) Effectiveness and Universality Validation: To verify the effectiveness and universality of RSUS, we conduct experiments with advanced semantic segmentation models on five different datasets. The selected semantic segmentation models include the following.

1) *SegFormer [65]:* This is an efficient and lightweight semantic segmentation network based on the transformer [54] architecture. Its lightweight decoder design enables us to evaluate the performance of RSUS to the greatest extent. All SegFormer models used in this article are of the SegFormer-B1 variant.
2) *SegNeXt [66]:* This is a classic lightweight convolutional neural network that realizes self-attention through basic

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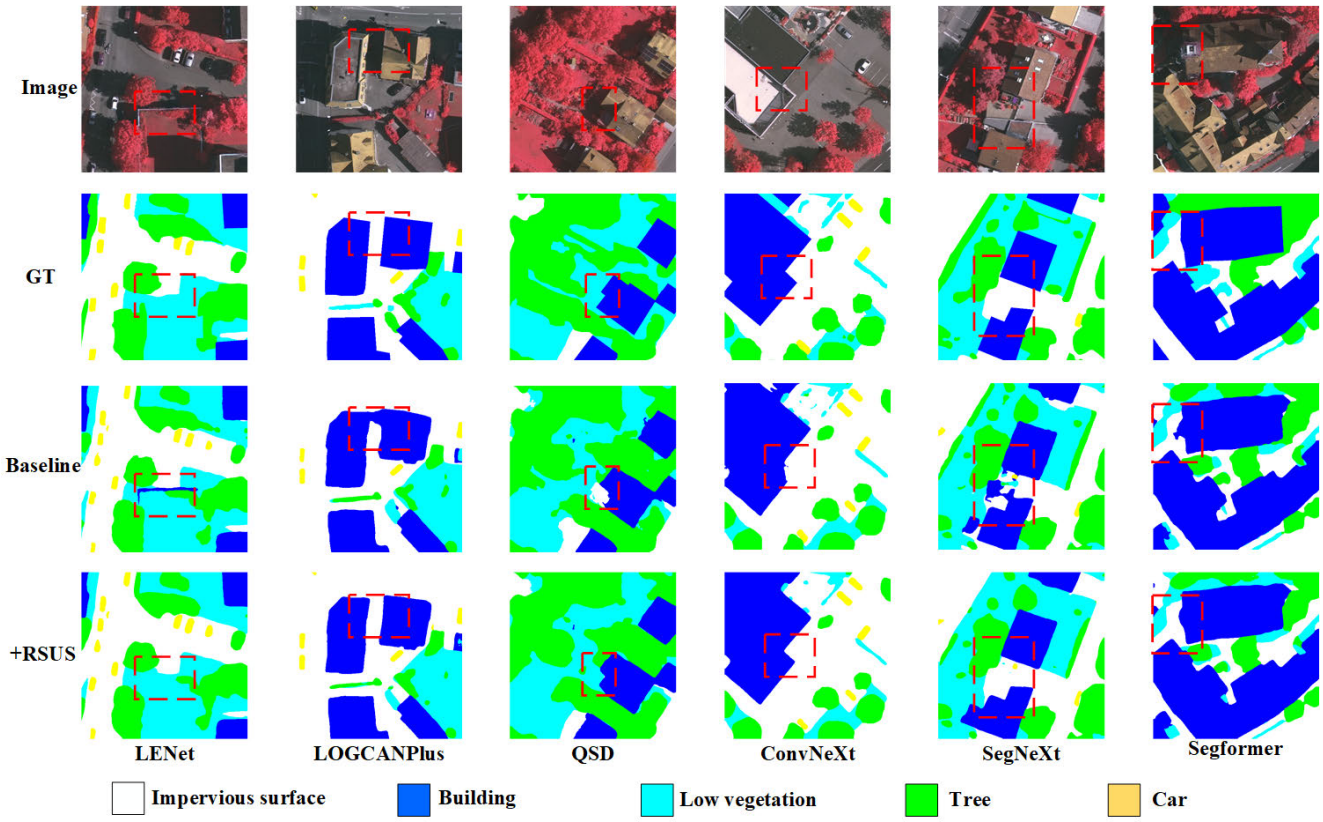


Fig. 7. Qualitative segmentation visualization on the ISPRS Vaihingen dataset.

convolutions. It efficiently replaces the attention mechanism of transformers with convolutional operations and has shown strong potential in RSI segmentation tasks. All SegNeXt models used in this article are of the SegNeXt-Tiny variant.

3) *ConvNeXt [67]:* This is an advanced convolution-based backbone network with excellent feature extraction capability. In this article, all ConvNeXt models are of the ConvNeXt-Tiny variant and are used together with UPerNet [68] as the decoder.
4) *QSD [69]:* This is an advanced network that combines spiking neural networks (SNNs) with transformers. It employs quantized spikes to drive the Transformer and has demonstrated great potential in RSI segmentation tasks.
5) *LOGCAN++ [13]:* This is an advanced semantic segmentation network designed for RSI land-cover classification tasks. By combining global class-aware and local class-aware strategies, it reduces background noise interference and alleviates the problem of large intraclass variance, making it one of the SOTA networks for RSI segmentation.
6) *LENet [9]:* This is an advanced semantic segmentation network designed for RSI segmentation tasks involving complex landforms. It effectively addresses intraclass inconsistency and interference from background features in RSI segmentation through an intraclass feature compensation mechanism and a global feature attention selection mechanism, making it one of the SOTA networks for RSI segmentation.

a) ISPRS Vaihingen and Potsdam datasets quantitative and qualitative analysis: The quantitative and qualitative analyses on the ISPRS Vaihingen and Potsdam datasets are presented in Table I and Figs. 7 and 8. From the quantitative results on these two datasets, it can be observed that, after integrating RSUS into different segmentation models, consistent performance improvements of varying degrees are achieved. The improvements are particularly more pronounced for relatively difficult categories such as low vegetation and cars. Meanwhile, the qualitative analysis based on the visual results shows that RSUS can significantly reduce intraclass feature loss and produce more complete intraclass information. It also yields more accurate segmentation boundaries for adjacent categories with similar features and preserves spatial structures involving multiple categories more completely. This can be attributed to the content-aware scheme of RSUS based on global feature aggregation, which enables the model to learn more complete global features of the same category. At the same time, CAIPE and AHSG further help RSUS constrain spatial structural displacement and provide accurate high-frequency boundary information for different categories.

b) LoveDA dataset quantitative and qualitative analysis: LoveDA is a classic RSI semantic segmentation dataset. In this article, we use the official online evaluation results provided by LoveDA for quantitative analysis, and we also conduct qualitative analysis by visualizing the prediction maps on the unlabeled test set. The results are shown in Fig. 9 and Table II. After incorporating RSUS, different models all achieve performance improvements to varying degrees, especially in the road category, which is relatively difficult to segment. For example,

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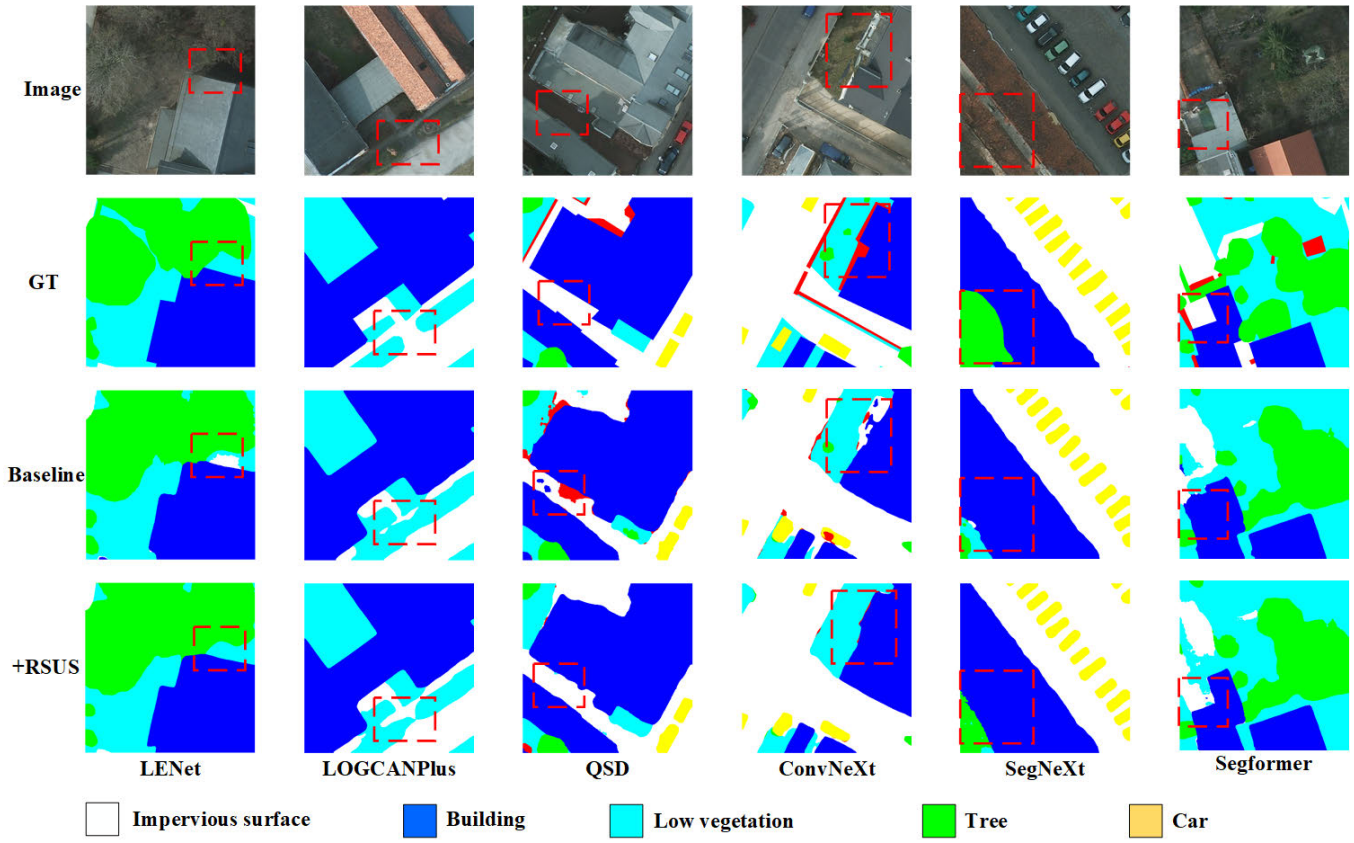


Fig. 8. Qualitative comparison of segmentation results for the ISPRS Potsdam dataset.

TABLE I

QUANTITATIVE RESULTS FOR UNIVERSALITY ANALYSIS ON THE ISPRS VAIHINGEN AND POTSDAM DATASETS (THE BETTER VALUES ARE MARKED IN BOLD, UNIT: %)

Methods	ISPRS Vaihingen								ISPRS Potsdam							
	Imp.	Bui.	Low.	Tree	Car	mIoU	mF1	OA	Imp.	Bui.	Low.	Tree	Car	mIoU	mF1	OA
Segformer	87.29	92.08	71.35	80.81	76.08	81.52	89.63	88.99	84.48	93.26	75.69	79.89	88.74	84.41	91.42	89.25
+RSUS	**87.33**	**92.76**	**72.87**	**81.01**	**79.63**	**82.72**	**90.39**	**90.62**	**85.25**	**93.65**	**76.03**	**80.37**	**89.41**	**84.94**	**91.73**	**89.54**
SegNeXt	86.44	92.47	70.78	79.78	78.18	81.55	89.65	89.59	**84.64**	93.29	75.53	80.15	88.7	84.47	91.45	89.30
+RSUS	**86.81**	**92.48**	**72.25**	**80.57**	**79.17**	**82.26**	**90.11**	**90.03**	84..43	**93.85**	**77.05**	**80.54**	**89.41**	**85.05**	**91.81**	**89.59**
ConvNeXt	87.27	92.94	72.00	80.34	80.43	82.59	90.30	90.60	87.91	93.90	78.06	80.55	92.04	86.49	92.63	91.36
+RSUS	**88.29**	**93.57**	**73.70**	**81.63**	**81.53**	**83.74**	**91.00**	**91.70**	**87.93**	**94.03**	**79.22**	**81.40**	**92.52**	**87.02**	**92.95**	**91.64**
QSD	86.72	92.00	71.31	80.04	77.50	81.51	89.64	90.63	86.69	92.42	76.78	78.95	90.48	84.87	91.80	89.70
+RSUS	**87.76**	**92.62**	**73.01**	**81.29**	**81.08**	**83.15**	**90.66**	**91.32**	**87.37**	**92.93**	**78.20**	**79.80**	**91.84**	**86.03**	**92.37**	**90.99**
LOGCAN++	88.01	93.48	73.81	**81.46**	82.22	83.79	91.16	91.32	87.24	93.57	77.93	80.36	91.70	86.16	92.45	91.06
+RSUS	**88.41**	**93.6**	**74.05**	81.44	**84.29**	**84.36**	**91.38**	**91.75**	**88.07**	**94.18**	**78.47**	**81.33**	**92.66**	**86.94**	**92.89**	**91.46**
LENet	87.85	92.90	73.68	81.45	79..02	82.98	90.55	91.47	87.34	94.05	74.23	**80.91**	90.24	85.35	91.94	89.78
+RSUS	**88.12**	**93.42**	**74.04**	**81.58**	**80.82**	**83.60**	**90.92**	**91.67**	**87.92**	**94.12**	**78.78**	79.78	**91.99**	**86.52**	**92.65**	**91.40**

in the SegFormer, ConvNeXt, and QSD models, the accuracy of the road category is significantly improved. This is because RSUS provides more complete long-range topological spatial structures, making long-distance category structures closer to the ground truth. For LENet, the incorporation of RSUS improves its segmentation performance on all categories to different extents, especially its ability to recognize categories in complex backgrounds. This is attributed to the global contextual feature aggregation capability of RSUS, which, together with topological structure constraints and the extraction of boundary high-frequency information, enables more accurate identification of ground object features in complex multicategory spaces. As shown in the figure, after integrating RSUS, different models exhibit clear improvements in large-scale category integrity, intraclass information consistency, and spatial structural accuracy. These results demonstrate that RSUS can effectively help different models improve their segmentation performance in RSI segmentation tasks with complex backgrounds.

c) UAVid dataset quantitative and qualitative analysis: The UAVid dataset is a challenging UAV-view dataset, and its quantitative and qualitative analyses are presented in Table III and Fig. 10. Since UAVid is captured from a top-view UAV perspective and contains more categories, it poses greater challenges for segmentation. After incorporating RSUS, all models exhibit noticeable performance improvements to varying degrees. For example, SegFormer shows a significant improvement in the recognition of the human category,

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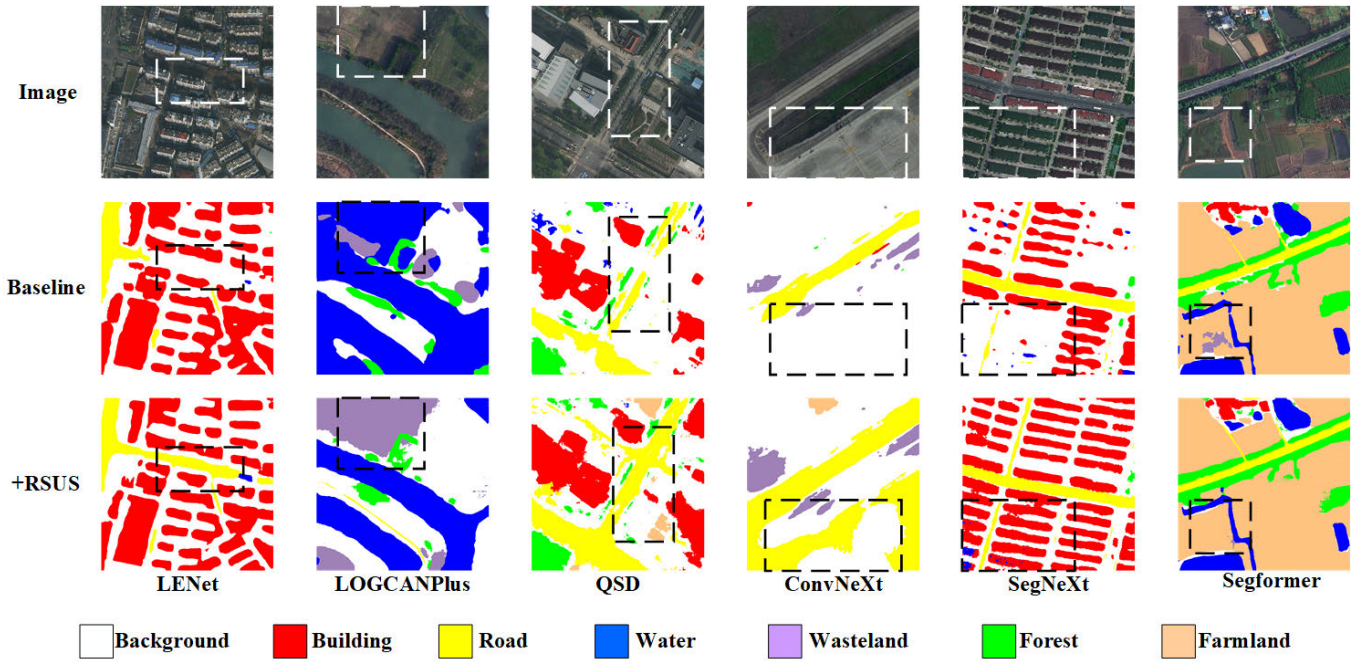


Fig. 9. Visualization of online test results on the LoveDA dataset, where the LoveDA test set lacks ground-truth labels.

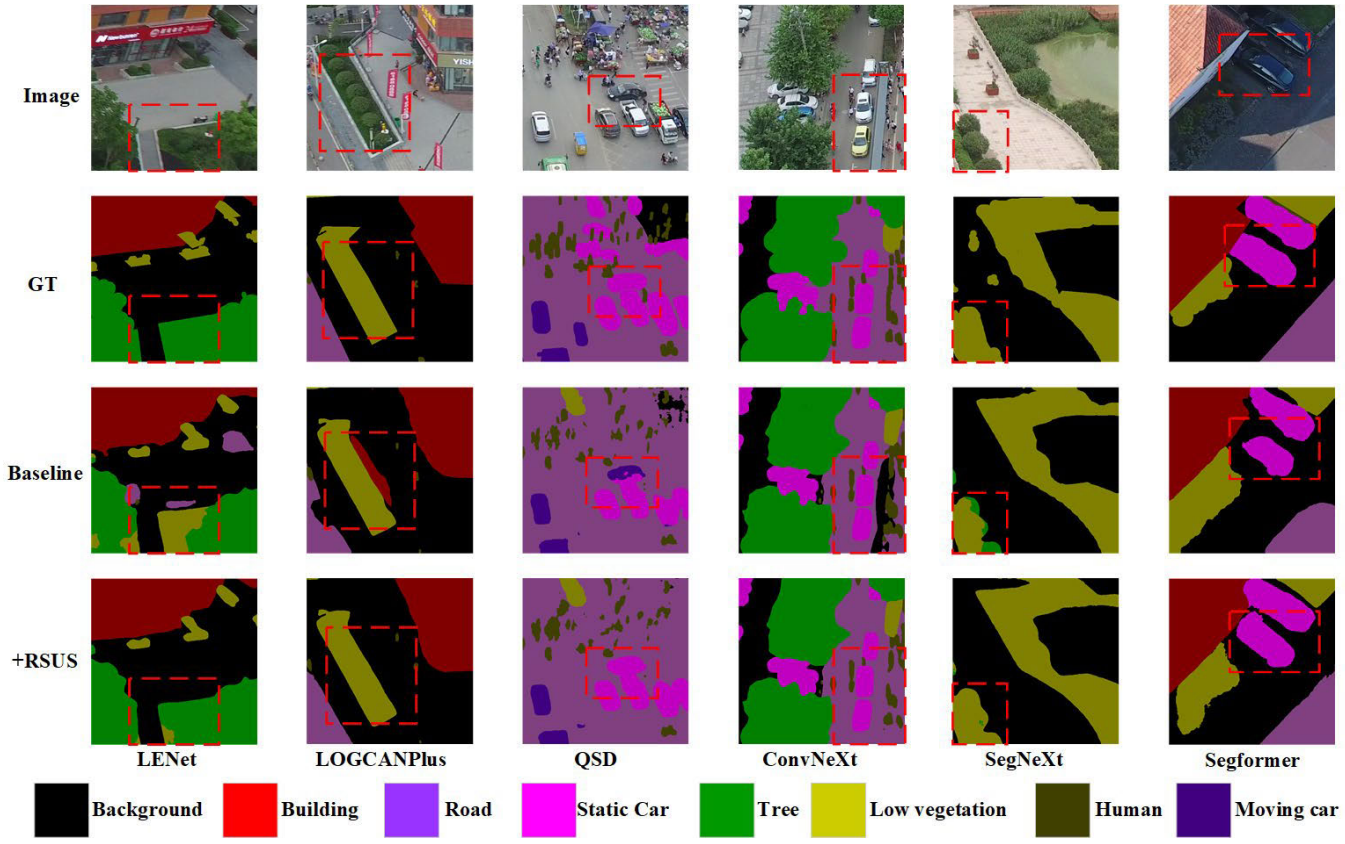


Fig. 10. Qualitative comparison of segmentation results on the UAVid UAV dataset.

because global contextual perception helps the model enhance the distinction between small categories and the background. Meanwhile, this capability also helps improve the integrity of large-scale target categories. For instance, in ConvNeXt, the recognition accuracy of the low vegetation category is significantly improved. In the visualization of segmentation results, for example, the misclassification problem for the same vehicle observed in QSD is effectively alleviated by the high-frequency feature extraction capability of RSUS, which helps the model produce more accurate category boundaries. In addition, in LOGCAN++, the phenomenon where background regions are misclassified as buildings leads to disordered

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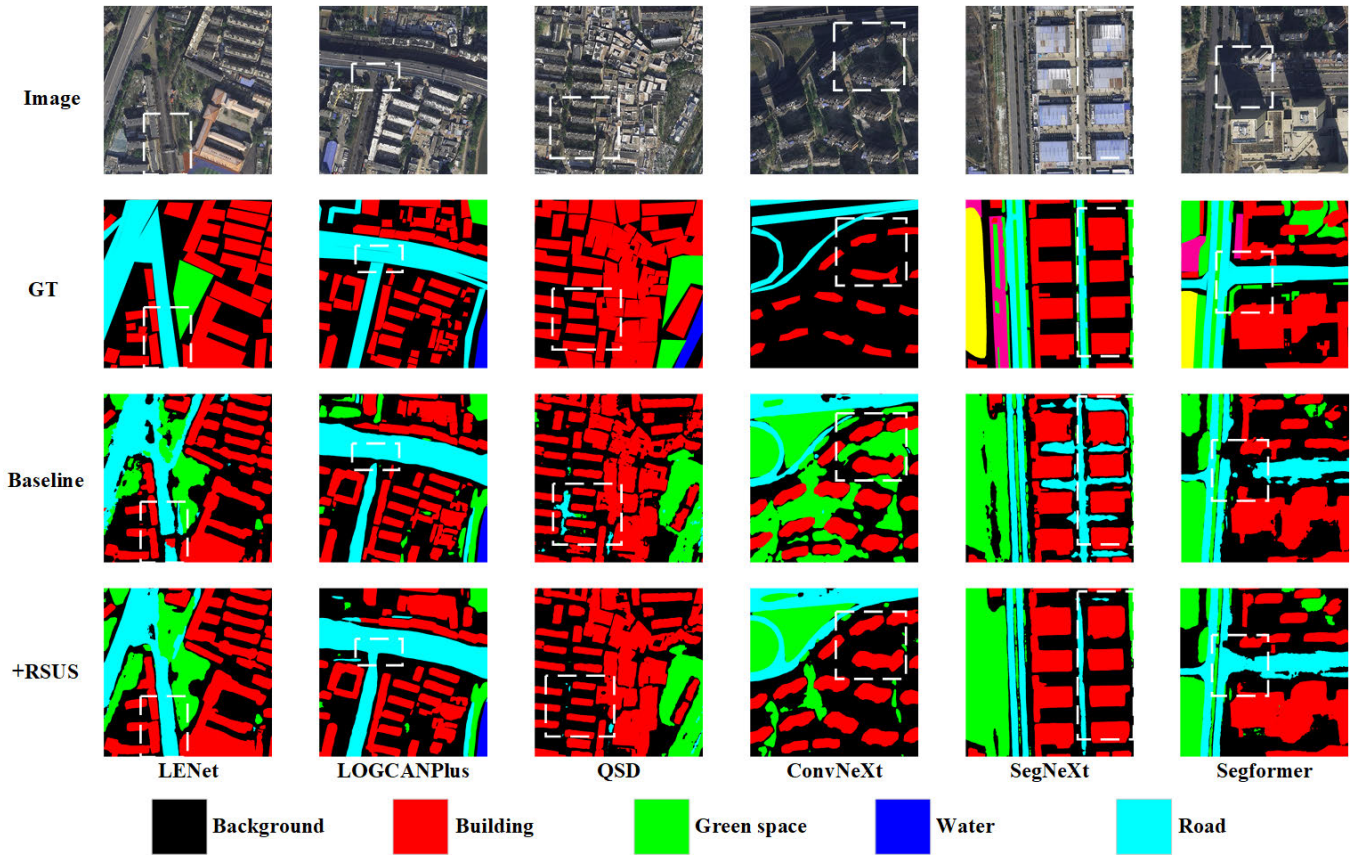


Fig. 11. Figure shows the visualization of the qualitative segmentation results of the Xining dataset.

TABLE II

ONLINE TEST RESULTS ON THE LOVEDA DATASET (ONLY IOU FOR EACH CATEGORY AND MIOU ARE RELEASED IN THE LOVEDA ONLINE TEST; THE BETTER VALUE OF EACH CATEGORY IS MARKED IN BOLD. UNIT: %)

Methods	Back.	Bui.	Road	Water	Bar.	For.	Agr.	mIoU
Segformer	44.79	56.8	54.12	**78.61**	17.48	**45.33**	60.39	51.07
+RSUS	**44.8**	**56.89**	**59.36**	77.66	**23.36**	44.31	**60.44**	**52.4**
SegNeXt	41	58.49	56.07	**79.37**	**16.35**	44.08	58.98	50.62
+RSUS	**43.36**	**59.91**	**58.84**	78.67	15.48	**46.23**	**59.47**	**51.71**
ConvNeXt	43.52	57.22	53.94	**80.04**	**20.59**	**47.37**	58.39	51.38
+RSUS	**46.53**	**57.42**	**58.12**	79.39	20.28	46.4	**60.68**	**52.7**
QSD	37.13	46.04	45.34	66.66	12.96	39.52	50.54	42.6
+RSUS	**37.24**	**51.24**	**47.54**	**75.45**	**16.32**	**40.21**	**61.39**	**47.05**
LOGCAN++	45.49	**60.16**	58.21	**78.37**	17.3	46.75	**64.55**	52.98
+RSUS	**46.13**	57.98	**58.46**	78.21	**24.54**	**48.38**	62.26	**53.71**
LENet	44.9	58.85	58.83	79.09	22.71	46.89	62.65	53.42
+RSUS	**48.02**	**60.48**	**58.86**	**80.69**	**25.06**	**48.69**	**63.75**	**55.08**

spatial topological relationships. By constraining spatial structural displacement and integrating global category features, RSUS effectively mitigates this misclassification problem. These results demonstrate that RSUS can still effectively help improve segmentation quality for low-altitude imagery segmentation under UAV-view scenarios.

d) Xining dataset quantitative and qualitative analysis: The Xining dataset is characterized by compact distributions of ground object categories and complex spatial structures, making it of great significance for investigating the capability of RSUS in spatial structure recovery and category information extraction. The quantitative and qualitative analyses on the Xining dataset are presented in Fig. 11 and Table IV. After incorporating RSUS, the overall segmentation accuracy of different models is improved to varying degrees. Since the Xining dataset contains a large number of wide roads extending over long spatial distances, all models show clear improvements in the road category after integrating RSUS. This fully demonstrates that the global feature aggregation capability and spatial structure constraint capability of RSUS can effectively assist models in recognizing large-scale targets. From the qualitative analysis of the segmentation results of different models, it can be observed that most models encounter problems such as road discontinuity or background misclassification. For example, both SegFormer and SegNeXt exhibit broken or misclassified road segments, while QSD and ConvNeXt suffer from background misclassification. After incorporating RSUS, these problems, including the loss of large-scale target integrity and the misclassification of targets with similar spectral features, are effectively alleviated. Therefore, it can be effectively demonstrated that, in complex land-cover classification tasks, RSUS can help improve the overall segmentation quality of different models.

2) Comparative Study With Advanced Upsampling Methods: To verify the superior adaptability of RSUS to RSI semantic segmentation tasks, we select SegFormer-B1 as the baseline model and compare RSUS with several advanced upsampling methods on the LoveDA test set. It should be noted that the LoveDA test set is evaluated through an online challenge platform, and therefore, no public ground-truth labels are available for the test set.

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TABLE III
QUANTITATIVE UNIVERSALITY ANALYSIS RESULTS ON THE UAVID UAV DATASET (THE BETTER VALUES ARE MARKED IN BOLD, UNIT: %)

Methods	Back.	Bui.	Road	Tree	Low.	MovCar.	StaCar.	Human.	mIoU	mF1	OA
Segformer	67.35	92.55	80.43	79.20	69.93	76.09	70.70	44.78	72.63	83.45	88.90
+RSUS	**69.12**	**93.08**	**81.82**	**79.86**	**70.76**	**78.19**	**74.18**	**47.63**	**74.33**	**84.66**	**89.53**
SegNeXt	68.31	92.84	81.57	79.46	70.04	76.63	72.47	**42.71**	73.00	83.61	89.22
+RSUS	**70.19**	**93.28**	**82.77**	**79.87**	**71.02**	**77.31**	**73.43**	41.12	**73.63**	**83.95**	**89.74**
ConvNeXt	69.26	93.14	**81.81**	79.51	70.40	77.96	74.77	49.30	74.52	84.84	89.46
+RSUS	**69.96**	**93.44**	81.58	**80.32**	**71.39**	**78.70**	**75.68**	**50.30**	**75.17**	**85.29**	**89.80**
QSD	67.45	92.48	80.24	78.91	68.99	73.48	69.71	42.77	71.75	82.79	88.73
+RSUS	**68.16**	**92.79**	**81.12**	**79.57**	**70.77**	**74.06**	**71.54**	**44.34**	**72.79**	**83.54**	**89.24**
LOGCAN++	69..63	93.31	81.43	79.74	70.70	**79.04**	74.81	49.58	74.78	85.02	89.59
+RSUS	**70.56**	**93.54**	**82.38**	**80.22**	**71.55**	78.78	**76.10**	**51.90**	**75.63**	**85.63**	**89.95**
LENet	69.20	93.11	81.64	79.74	70.53	77.22	73.06	46.40	73.86	84.32	89.47
+RSUS	**70.00**	**93.32**	**82.46**	**80.15**	**71.29**	**78.61**	**75.67**	**46.69**	**74.77**	**84.92**	**89.91**

TABLE IV
QUANTITATIVE UNIVERSALITY ANALYSIS RESULTS ON THE XINING DATASET (THE BETTER VALUE OF EACH CATEGORY IS MARKED IN BOLD. UNIT: %)

Methods	Back.	Bui.	Gre.	Water	Road	mIoU	mF1	OA
Segformer	49.77	**71.22**	**62.22**	79.18	48.97	62.27	76.10	73.79
+RSUS	**50.31**	70.95	61.67	**79.63**	**50.05**	**62.52**	**76.32**	**73.85**
SegNeXt	47.76	70.67	59.49	77.88	50.37	61.23	75.32	72.59
+RSUS	**50.13**	**72.33**	**61.16**	**79.30**	**50.91**	**62.77**	**76.51**	**74.04**
ConvNeXt	49.10	71.77	**62.49**	**79.54**	48.96	62.37	76.14	73.83
+RSUS	**50.93**	**72.69**	62.27	79.10	**50.72**	**63.14**	**76.81**	**74.56**
QSD	47.69	69.76	57.97	77.68	47.95	60.21	74.48	71.73
+RSUS	**50.10**	**69.89**	**60.51**	**79.81**	**49.00**	**61.86**	**75.79**	**73.15**
LOGCAN++	49.92	71.8	**61.54**	79.54	50.61	62.68	76.44	73.92
+RSUS	**51.22**	**72.69**	61.43	**81.60**	**51.94**	**63.77**	**77.25**	**74.53**
LENet	**49.72**	70.49	**61.82**	79.85	50.42	62.46	76.27	73.57
+RSUS	49.41	**73.38**	60.20	**80.59**	**51.62**	**63.04**	**76.66**	**74.11**

TABLE V
COMPARATIVE QUANTITATIVE RESULTS OF DIFFERENT SOTA UPSAMPLING METHODS (THE BEST VALUE OF EACH CATEGORY IS MARKED IN BOLD. UNIT: %)

Methods	Back.	Bui.	Road	Water	Bar.	For.	Agr.	mIoU
Bilinear (Baseline)	44.79	56.80	54.12	78.61	17.48	45.33	60.39	51.07
CARAFE++	44.38	55.39	57.15	78.75	18.31	46.57	61.36	51.70
DySample	**45.28**	53.99	58.86	**79.28**	17.15	46.77	63.09	52.09
FADE	43.58	55.03	57.79	77.10	16.08	**46.98**	62.25	51.26
Converse	44.07	54.88	58.43	78.14	18.66	46.21	62.04	51.77
Freqfusion	44.63	54.80	58.29	77.90	19.61	45.85	**63.53**	52.09
RSUS(Ours)	44.80	**56.89**	**59.36**	77.66	**23.36**	44.31	60.44	**52.40**

The advanced upsampling methods included in the comparative experiments are as follows.

1) *CARAFE++ [15]:* A general, efficient, and lightweight content-aware reassembly operator that performs content-aware reconstruction for different instances through adaptive kernels.
2) *DySample [19]:* A point-sampling-based dynamic convolution upsampling scheme that is lightweight and efficient and has demonstrated excellent performance in semantic segmentation tasks.
3) *FADE [21]:* An improved and efficient content-aware reassembly operator that can effectively control the stationary-point contributions of encoder–decoder features through a half-shift operator, thereby enabling selective feature transfer to compensate for upsampling details.
4) *Converse [18]:* A novel deconvolution operator that can explicitly capture spatial dependencies. It can not only serve efficiently as an upsampling layer in semantic segmentation networks, but has also demonstrated effectiveness in tasks such as Gaussian denoising and super-resolution.
5) *FreqFusion [25]:* An efficient feature fusion method designed to address intraclass inconsistency and boundary displacement. It adaptively smooths high-level features through spatially varying low-pass filtering, replaces inconsistent features in high-level representations with resampled nearby class-consistent features, and enhances the high-frequency components of low-level features, thereby efficiently achieving both feature fusion and feature upsampling.

a) Quantitative and qualitative analysis: The quantitative results are presented in Table V. Different advanced upsampling methods achieve performance improvements to varying degrees over the Baseline (Bilinear), among which RSUS surpasses the current SOTA FreqFusion by 0.31 in mIoU. In addition, for the relatively difficult road and barren categories, RSUS achieves the highest per-class IoUs of 59.36 and 23.36, respectively. This can be attributed to the sensitivity of RSUS to long-range features and its ability to capture global structural information.

As shown in Fig. 12, the prediction visualizations of different methods on the LoveDA test set further demonstrate that, compared with the Baseline and other advanced methods, RSUS exhibits better capability in preserving the integrity of large-scale structures, while also achieving better intraclass feature consistency. In particular, when dealing with two adjacent large-scale categories with similar spectral characteristics or categories extending over long distances, RSUS can better help the model recover intraclass features and global structural details.

b) Floating-point numbers and parameters and computational efficiency analysis: As listed in Table VI, we compare the growth of FLOPs and parameter quantities between the proposed RSUS and other SOTA upsampling methods. Although RSUS exhibits the largest parameter increment among all competitors, its FLOPs overhead remains moderate. Further combining the metrics of inference speed and peak GPU memory occupation, RSUS achieves competitive

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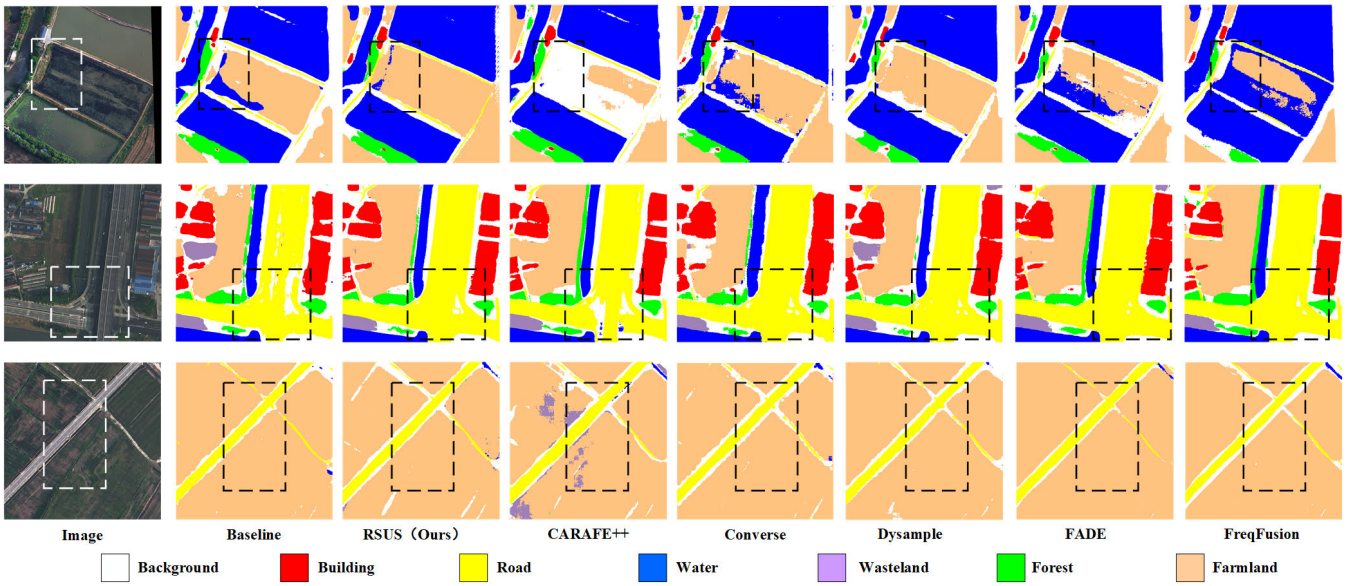


Fig. 12. Qualitative comparison of online prediction results from various advanced upsampling methods for the LoveDA dataset.

TABLE VI
COMPARISONS OF FLOATING-POINT OPERATIONS, PARAMETER COUNTS, INFERENCE SPEED, AND GPU MEMORY CONSUMPTION ACROSS DIFFERENT ADVANCED UPSAMPLING METHODS

Methods	FLOPs (G)	Params (M)	Latency (ms/img)	GPU memory (MiB)
Bilinear	86.67	13.68	7.25	347.84
CARAFE++	89.59	14.94	31.79	2416.42
DySample	87.01	13.85	15.44	816.80
FADE	96.90	13.73	103.66	3635.02
Converse	86.46	13.70	107.00	1740.54
Freqfusion	92.69	13.89	54.08	10195.31
RSUS(Ours)	93.89	17.04	24.51	911.53

efficiency, with its inference latency and peak GPU memory usage only surpassed by bilinear interpolation and DySample. This advantage mainly stems from our adoption of standard Einstein summation, which avoids the intermediate tensor expansion adopted by traditional content-aware upsampling schemes and substantially cuts practical hardware resource overhead. Collectively, these observations demonstrate that RSUS delivers prominent segmentation performance gains with a tolerable computational cost increase.

3) t-SNE-Based Class Feature Distribution and Topological Analysis: In this section, class feature distribution and topological analysis are jointly employed to investigate the improvement brought by RSUS in category clustering capability and the structural integrity of topological space. Specifically, t-SNE [70] is used to visualize the last-layer features output by the model for class feature distribution analysis, while the PH introduced in Section IV-B is used to visualize the capability of RSUS in improving connected components and loop structures. In all experiments of this section, ConvNeXt is adopted as the baseline model, and the last-layer features are extracted and visualized on the validation set of the Vaihingen dataset. As shown in Fig. 13, we use t-SNE to visualize the feature distribution of ConvNeXt on the Vaihingen dataset. Before incorporating RSUS, the baseline model exhibits relatively weak discrimination between some similar categories, such as impervious surfaces and low vegetation. After incorporating RSUS, the model improves the separability of these similar categories. This can be attributed to the sensitivity of RSUS to frequency-domain boundary information, which enhances the discrimination between different categories. Meanwhile, the capability of RSUS to constrain topological spatial structural displacement further strengthens the clustering ability of the original categories, such as buildings and trees.

As shown in Figs. 14 and 15, they present the 0-D homology and 1-D homology of ConvNeXt before and after incorporating RSUS, respectively. 0-D homology reflects the largest connected components that can persist across different scales, such as long-distance roads and rivers. In the baseline model, as the dilation radius increases, the impervious surface and building categories exhibit discrete abnormal connected points, which are often manifested as intraclass inconsistency or boundary overflow in the segmentation results. In contrast, for the low vegetation and tree categories, which possess long-range connectivity, the overall connected points vanish earlier, which is often reflected as the loss of intraclass information in the segmentation results. After incorporating RSUS, benefiting from its global feature aggregation and frequency-domain boundary perception capabilities, the connected points of categories with large-scale topological connectivity, such as impervious surfaces and buildings, become more continuous overall. Moreover, the connected points of the other categories can also be learned by the model under larger dilation radii. Therefore, RSUS can effectively help the model preserve the category connectivity of larger scale features during the upsampling process.

Meanwhile, 1-D homology can be used to analyze the model's ability to recognize loop structures under different dilation radii. If the distance of a loop point from the diagonal

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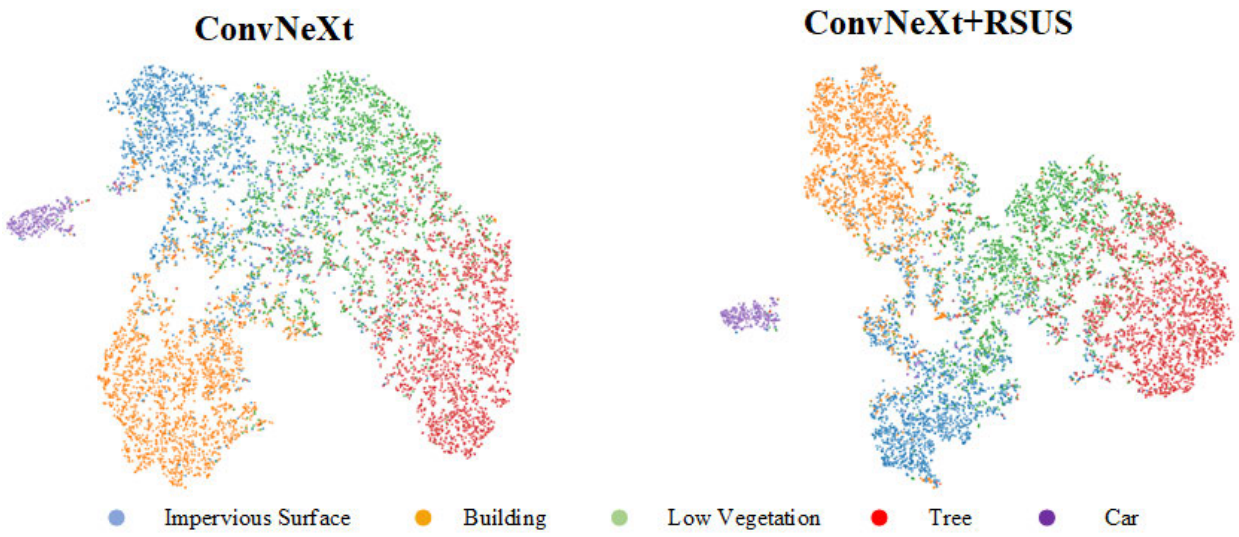


Fig. 13. t-SNE feature distribution of ConvNeXt on the Vaihingen dataset before and after embedding RSUS.

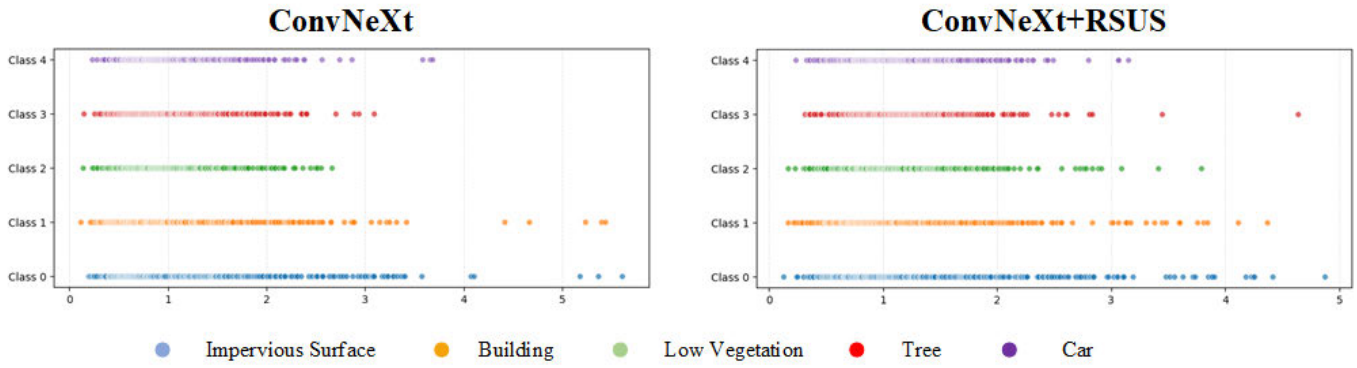


Fig. 14. 0-D PH Distributions of ConvNeXt with and without RSUS.

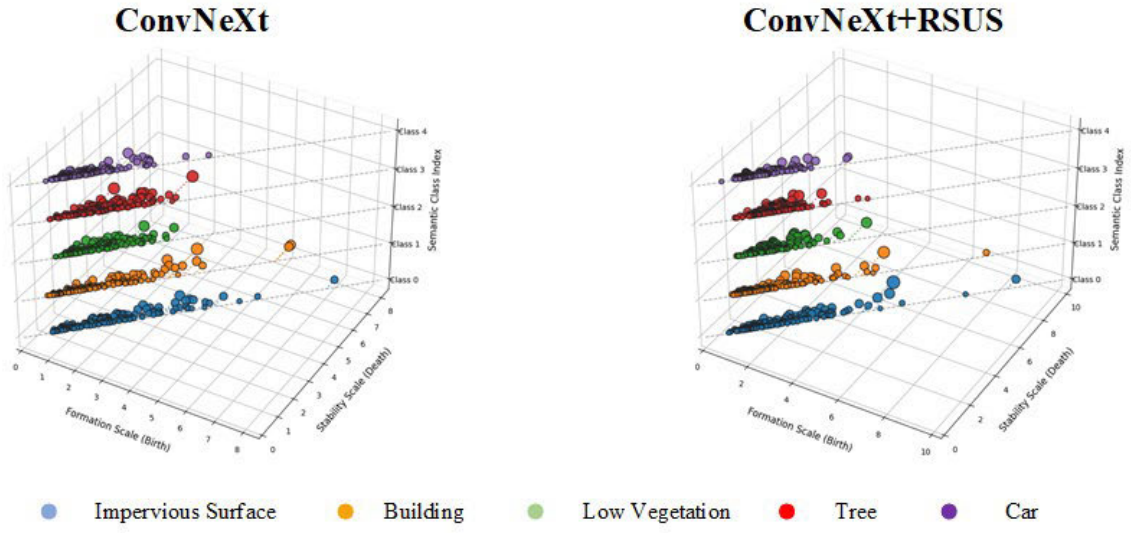


Fig. 15. 1-D PH Distributions of ConvNeXt with and without RSUS.

(birth-death line) is larger, it indicates that the loop closes later, meaning that the model can still recognize the topological structure at a larger scale. Compared with the baseline model, the capability of RSUS to constrain topological structural displacement and aggregate global contextual information helps the model recognize more complete large-scale features, such as impervious surfaces and buildings, which often still preserve complete topological structures at large scales.

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TABLE VII
QUANTITATIVE TOPOLOGICAL EVALUATION OF RSUS AND THE BASELINE METHOD FOR STRUCTURE-PRESERVING UPSAMPLING

Metric	Baseline	RSUS	Change (%)
H0 Bottleneck	1.25	1.18	-5.88
H1 Bottleneck	0.46	0.43	-6.06
H0 Wasserstein	61.36	35.55	-42.05
H1 Wasserstein	17.36	12.79	-26.31
Betti0 error	3.00	2.80	-6.67
Betti1 error	2.20	1.60	-27.27

Therefore, RSUS can effectively recover complete large-scale topological structures during the upsampling process in RSI segmentation tasks.

To further support the qualitative PH visualization, we report quantitative topology-consistency metrics in Table VII. Compared with the baseline, RSUS achieves lower bottleneck distances in both 0-D and 1-D homology, with reductions of 5.88% and 6.06%, respectively. This indicates that RSUS can reduce the maximum topological discrepancy between the predicted and ground-truth persistence diagrams, thereby constraining the structural displacement introduced during upsampling.

In addition, RSUS significantly reduces the Wasserstein distances, with H0 Wasserstein decreasing from 61.36 to 35.55 and H1 Wasserstein decreasing from 17.36 to 12.79, corresponding to reductions of 42.05% and 26.31%, respectively. Since Wasserstein distance measures the accumulated discrepancy of persistence diagrams, these results demonstrate that RSUS improves the overall consistency of connected components and loop structures with the ground truth. Furthermore, RSUS reduces Betti0 error from 3.0 to 2.8 and Betti1 error from 2.2 to 1.6, indicating that the predicted number of connected components and loop structures is closer to the ground truth. These quantitative results confirm that RSUS does not merely enhance topological responses but more effectively preserves structurally faithful topology and suppresses topological displacement during the upsampling process (since the persistence diagrams are computed from class-wise feature point clouds, the Bottleneck and Wasserstein distances are measured in the feature-space distance scale).

4) Ablation Study: In this section, ablation experiments are conducted on RSUS to demonstrate the contribution of each component in the upsampling process. SegFormer-B1 and ConvNeXt-Tiny are adopted as experimental networks, and experiments are performed on the ISPRS Vaihingen, ISPRS Potsdam, and LoveDA datasets to validate the general applicability of RSUS as a unified upsampling method. We design four ablation configurations and two baseline settings as follows.

a) Two baseline configurations: We adopt bilinear interpolation and CARAFE++ as baseline methods, where bilinear interpolation represents traditional interpolation-based feature upsampling, and CARAFE++ represents content-aware upsampling methods.

b) Four ablation configurations:

1) *GCVA Only:* In this setting, only GCVA is retained. The global context-aware kernel serves as the core component of RSUS. It aggregates global contextual information to help the prediction kernel capture more complete category semantics and expand its receptive field. In addition, we optimize the content-aware upsampling process by replacing the intermediate tensor expansion in the original content-aware kernel with a standard Einstein summation formulation, thereby improving computational compactness. This setting is used to evaluate the contribution of global context aggregation to prediction-kernel generation, as well as the effect of the optimized content-aware reconstruction;
2) *GCVA and CAIPE:* Based on setting (1), CAIPE is further introduced to impose topological spatial constraints. This component integrates a content-aware kernel with global contextual aggregation and a cross-scale anisotropic positional encoding mechanism, which captures topological geometric structures. This setting is used to evaluate the performance gain brought by incorporating spatial structural priors;
3) *GCVA and AHSG:* Based on setting (1), AHSG is introduced to enable adaptive extraction of high-frequency boundary information. This setting is designed to investigate the contribution of frequency-aware boundary enhancement to upsampling performance;
4) *Full RSUS:* This setting integrates GCVA, CAIPE, and AHSG to form the complete RSUS framework. The results of this setting demonstrate the overall performance improvement achieved through the joint effect of all components.

The ablation results are reported in Table VIII. First, compared with CARAFE++, setting (1), which retains GCVA and adopts Einstein-summation-based content-aware reconstruction, achieves better performance on most datasets and backbones. For SegFormer, setting (1) improves mIoU from 81.42 to 81.95 on Vaihingen and from 51.70 to 51.77 on LoveDA. For ConvNeXt, it improves mIoU from 82.57 to 82.99 on Vaihingen, from 84.96 to 85.44 on Potsdam, and from 51.33 to 51.72 on LoveDA. These results indicate that incorporating global contextual priors into the prediction-kernel generation process is more effective than conventional local content-aware reassembly. Moreover, the Einstein summation formulation eliminates explicit tensor expansion, resulting in a more compact and efficient reconstruction process.

However, setting (1) still underperforms bilinear interpolation in certain cases, particularly on the Potsdam dataset. This suggests that global context-guided content-aware reconstruction alone is insufficient to fully address structural displacement and boundary recovery challenges in remote sensing segmentation. Therefore, additional spatial constraints and frequency-aware boundary modeling are necessary.

After introducing CAIPE, setting (2) consistently outperforms setting (1). For example, on ConvNeXt, mIoU increases from 82.99 to 83.33 on Vaihingen, from 85.44 to 86.20 on Potsdam, and from 51.72 to 52.29 on LoveDA. This demonstrates that anisotropic implicit positional encoding effectively provides cross-scale geometric priors, reducing spatial misalignment and improving structural consistency.

Setting (3), which introduces AHSG on top of GCVA, also yields consistent improvements over setting (1). For

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TABLE VIII
ABLATION STUDY OF RSUS COMPONENTS

Methods	Sequence	Vaihingen			Potsdam			LoveDA
		mIoU	mF1	OA	mIoU	mF1	OA	mIoU
Segformer	Baseline	81.52	89.63	88.99	84.41	91.42	89.25	51.07
	CARAFE++	81.42	89.58	88.97	83.65	90.00	88.05	51.70
	(1)	81.95(↓0.77)	89.69(↓0.7)	89.89(↓0.73)	84.09(↓0.85)	90.52(↓1.21)	88.81(↓0.73)	51.77(↓0.63)
	(2)	82.23(↓0.49)	90.02(↓0.37)	90.16(↓0.46)	84.3(↓0.64)	91.07(↓0.66)	89.19(↓0.35)	51.92(↓0.48)
	(3)	82.19(↓0.53)	89.80(↓0.59)	90.1(↓0.52)	84.45(↓0.49)	91.35(↓0.38)	89.27(↓0.27)	52.04(↓0.36)
	(4)	82.72	90.39	90.62	84.94	91.73	89.54	52.40
ConvNeXt	Baseline	82.59	90.30	90.60	86.49	92.63	91.36	51.38
	CARAFE++	82.57	90.28	90.63	84.96	91.75	89.92	51.33
	(1)	82.99(↓0.75)	90.51(↓0.49)	90.85(↓0.85)	85.44(↓1.58)	92.03(↓0.92)	90.59(↓1.05)	51.72(↓0.98)
	(2)	83.33(↓0.41)	90.67(↓0.33)	91(↓0.70)	86.2(↓0.82)	92.39(↓0.56)	91.03(↓0.61)	52.29(↓0.41)
	(3)	83.48(↓0.26)	90.84(↓0.16)	91.11(↓0.59)	86.59(↓0.43)	92.71(↓0.24)	91.4(↓0.24)	52.10(↓0.60)
	(4)	83.74	91.00	91.70	87.02	92.95	91.64	52.70

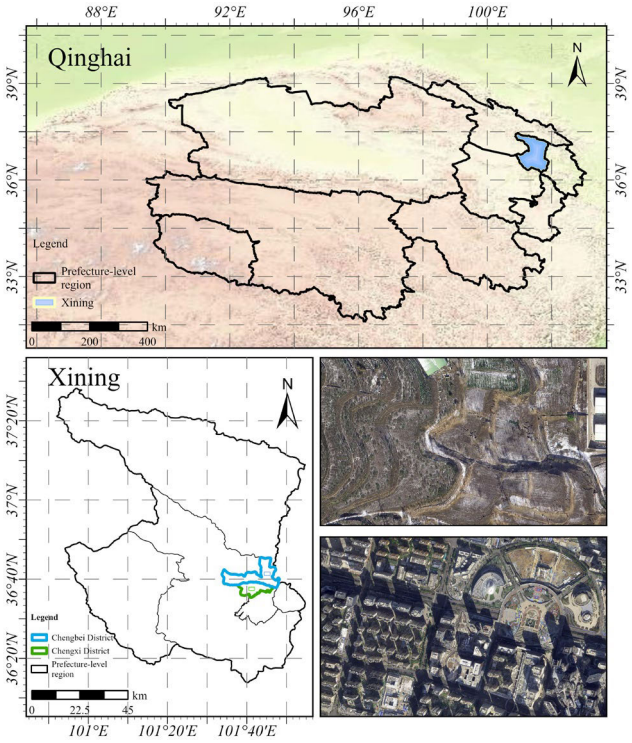


Fig. 16. Schematic map of the study area for the Xining dataset.

SegFormer, mIoU increases from 81.95 to 82.19 on Vaihingen, from 84.09 to 84.45 on Potsdam, and from 51.77 to 52.04 on LoveDA. For ConvNeXt, it further improves mIoU from 82.99 to 83.48 on Vaihingen and from 85.44 to 86.59 on Potsdam. These results verify that adaptive high-frequency structural gating enhances boundary-sensitive responses and improves local structural reconstruction during upsampling.

Finally, the full RSUS [setting (4)] achieves the best overall performance across all datasets and both backbones. Compared with setting (1), it improves SegFormer by 0.77, 0.85, and 0.63 mIoU on Vaihingen, Potsdam, and LoveDA, respectively. For ConvNeXt, the corresponding improvements are 0.75, 1.58, and 0.98 mIoU. Compared with settings (2) and (3), the full model still achieves further gains, indicating that CAIPE and AHSG are complementary. Specifically, CAIPE primarily improves cross-scale spatial alignment, while AHSG enhances frequency-domain structural sensitivity. Their integration with GCVA enables RSUS to simultaneously improve semantic consistency, structural alignment, and boundary reconstruction.

Overall, the ablation study demonstrates that each component contributes from a distinct perspective. GCVA enhances content-aware upsampling through global contextual modeling; CAIPE improves structural alignment via anisotropic geometric priors; and AHSG strengthens boundary and local structural details through frequency-aware gating. The full RSUS achieves the best performance, confirming the necessity and complementarity of all three components.

V. CONCLUSION

In this article, we conduct a detailed analysis of the challenges faced by RSI segmentation tasks, including complex background features, large intraclass variance, and complicated spatial structure distributions. Under this background, we propose that designing task-specific upsampling schemes for RSI segmentation can further unleash the potential of RSI semantic segmentation models. The proposed RSUS is evaluated through universality and robustness experiments on multiple models across five different datasets, and is further compared with other advanced upsampling methods on a public dataset under the same benchmark setting. The experimental results demonstrate that RSUS not only effectively improves model performance on RSI segmentation tasks but also achieves the best performance gains compared with SOTA methods. This is because RSUS is built upon content awareness guided by global feature aggregation, while further integrating cross-scale anisotropic spatial encoding and adaptive high-frequency structural gating, thereby effectively addressing the problems of structural misalignment and category feature loss after upsampling in RSI segmentation tasks. This work is intended to encourage more researchers to explore customized upsampling schemes for RSI tasks, so as to address the challenges of target recognition in the field of Earth observation.

APPENDIX

In this Appendix, we describe in detail the data source, a notation protocol, class definition, and disclosure of the Xining dataset exposed in this article.

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As shown in Fig. 16, the data acquisition area is the main urban area of Xining, Qinghai, China, which is located in the eastern part of Qinghai and the northeastern part of the Qinghai-Tibet Plateau. The total area of the region is about 7660 km^2, of which the urban area is about 476.5 km^2. The geographical coordinates range from 100°52′ to 101°54′E and from 36°13′ to 37°28′N. BJ-3 satellite imagery was used in the study area, with a spatial resolution of 0.3 m and containing three bands of Red (R), Green (G), and Blue (B).

The total size of the Xining dataset is 72 RSIs covering 5400 × 3631 or 2694 × 5673. Referring to the construction of the ISPRS Vaihingen and Potsdam dataset and LoveDA dataset, we divided the complete 72 RSIs into a training set, test set, and unlabeled generalization set. The complete training set consists of 34 images, the complete test set consists of 332 images, and the unlabeled generalization set consists of six images. Referring to the classification method of public datasets and combining the land cover category characteristics of this region, we used three annotators to independently annotate the same image and cross-compound to complete the annotation of the Xining dataset. After cross-checking the results, we also discussed and rejudged the disagreement areas and finally completed the annotation. Annotators used the same standard to classify the images of the region into seven different categories: background, building, green space, water, road, barren land, and farmland. In addition, according to the distribution characteristics of cities in the Xining region, we divided the complete dataset into two subdata: city and complex landform (the bottom right two parts in Fig. 16, the top half is part of complex landform subdata, and the bottom half is part of city). The city subdataset contains 23 images in the training set and 16 images in the test set. There are 11 images in the training set and 16 images in the test set in the complex landform subdataset. The unlabeled generalization set is independent of the urban and complex landform subdatasets.

Compared with the public datasets, the buildings in the Xining dataset are more densely distributed and compact, and there is obvious boundary blur and category confusion. The Xining dataset has strong topology scene complexity and challenging segmentation, which can effectively verify the performance of RSUS. The experiment in this article uses the city subdataset in the Xining dataset, and combines the data distribution law to select the first five categories for evaluation. The complete dataset can be found at https://github.com/Ronin-711/RSUS

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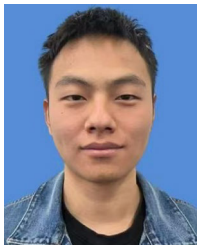
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